



Study on available traceability systems and tools in the Lao context

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Acronyms

BMZ	German Federal Ministry of Economic Cooperation and Development
COC	Chain of Custody
DAFO	District Agriculture and Forestry Office
DIMEX	Department of Import and Export
DLF	Department of Livestock and Fisheries
DOF	Department of Forestry
DOFI	Department of Forest Inspection
DOFT	Department of Foreign Trade
EUDR	European Union Deforestation Regulation
FAO	Forest and Agriculture Organisation of the United Nations
FLEGT	Forest Law Enforcement, Governance and Trade
FM	Forest Management
FSC	Forest Stewardship Council
GIZ	Deutsche Gesellschaft für Internationale Zusammenarbeit GmbH
GMP	Good Manufacturing Practices
GPS	Global Positioning System
HS Codes	Harmonised System Codes
IMS	Information Management System
ITC	International Trade Centre
LCA	Lao Coffee Association
LNCCI	Lao National Chamber of Commerce and Industry
LRA	Lao Rubber Association
MAF	Ministry of Agriculture and Forestry
MOF	Ministry of Finance
MOIC	Ministry of Industry and Commerce
MOLSW	Ministry of Labour and Social Welfare
MONRE	Ministry of Natural Resources and Environment
NSDEP	National Socio-Economic Development Plan
NTFPs	Non-Timber Forest Products
PAFO	Provincial Agriculture and Forestry Office
PEFC	Program for the Endorsement of Forest Certification
PFA	Production Forest Area

POFI	Provincial Office of Forest Inspection
SCR	Standard Chinese Rubber
SMEs	Small and Medium-sized Enterprises
SVR	Standard Vietnamese Rubber
TLAS	Timber Legality Assurance System
TLD	Timber Legality Definition
VPA	Voluntary Partnership Agreement

Executive Summary

The EU Deforestation Regulation (EUDR) entered into force in 2023, aims to combat global deforestation and forest degradation by ensuring that commodities and products placed on the EU market or exported from there are produced legally deforestation-free. It applies to key commodities such as soy, palm oil, coffee, cocoa, rubber, cattle and wood¹, as well as derived products. A central pillar of the EUDR is the traceability requirement, mandating operators who place relevant products on the market or export from there to provide geolocation data for production areas and ensuring due diligence to assess and mitigate potential risks. Traceability schemes and tools can be very useful tools for gathering relevant information, especially for assuring that the respective commodities are deforestation-free and in the case of timber, also forest degradation-free.

This study was implemented through a desk review for the commodities relevant for Lao PDR (coffee, rubber, cattle and timber) and interviews with stakeholders from the coffee, rubber and cattle sectors, including on-site visits of Lao operators (cooperatives, companies), serving as case studies. It provides an overview on the supply chains of the selected sectors in Lao PDR, describes the status quo of traceability within the country, outlines existing traceability schemes and tools and assesses their alignment with the EUDR traceability requirements. Based on this analysis, opportunities, benefits and challenges for supply chain actors to ensure deforestation-free commodity production are identified and recommendations for improvement of traceability schemes with regards to EUDR readiness are outlined.

Many respondents to the survey reported using some form of traceability system; these systems generally cover only partial stages of the supply chain rather than allowing for tracing back of a product to the specific plot of land where the commodity was produced. When compared to the traceability requirements outlined in Article 9 of the EUDR, most stakeholders were only able to provide limited information, primarily focusing on selected criteria such as product quantities or buyer details. Traceability systems in the rubber and cattle sector are very basic or inexistent. For the coffee industry, they are found to be slightly better established in general and in some cases in an advanced stage, due to the adoption of certification schemes and a focus on niche markets for sustainably produced coffee in overseas countries, including EU member states. Yet, in these systems, traceability is limited to the origin of farmers and not the geolocation of plots. However, challenges remain also in the coffee sector due to complex and fragmented supply chain structures, lack of knowledge and capacity as well as limited access to financial and technological resources. Traceability is a key element of the Lao Timber Legality Assurance System (TLAS) which is currently further developed as a national mandatory traceability system and builds on the achievements of the TLAS developed under the terminated Lao-EU FLEGT-VPA process. The process included an in-depth analysis of the legislation on supply chain management of timber sources, wood processing and trade, followed by legislative amendments, not yet concluded for all timber sources and still incomplete for wood processing. Traceability to the plot is legally required for natural timber from Production Forest areas, conversion areas and plantations, but not yet aligned with EUDR geolocation requirements regarding size of the area and digital geolocation formats.

The study concludes by providing a set of recommendations to enhance EUDR readiness with regards to traceability in the Lao supply chains covered by the analysis. Drawing on lessons from the VPA process and the TLAS development, key stakeholders, including government, private sector, and civil society, must be actively engaged in developing and implementing traceability systems across sectors. Technical and financial support is essential to develop a digital traceability system that empowers affected stakeholders along the supply chain to record and track product information. Solutions should be context-appropriate, with cost-effective systems for remote areas, and support for infrastructure like IT equipment for

¹ This study uses the term “timber” instead of “wood” used in the EUDR

smallholders. In addition, a national legal framework, in line with national green-growth strategies and socio-economic development and considering EUDR, is beneficial for ensuring consistency in traceability. Given the currently limited capacities and access to skills based on development programs of producers as well as requests for support articulated during the interviews with the private sector, training for Lao producers, particularly smallholders, on collecting traceability data and using digital tools is vital, as is the creation of a manual or app to guide them through the process and enhance EUDR readiness.

1. Introduction

Almost 90% of worldwide deforestation is the result of agricultural expansion. A significant share of the agricultural commodities driving deforestation are traded internationally. The consumption of the European Union is responsible for about 10% of the worldwide deforestation, particularly resulting from commodities such as soya and palm oil (European Parliament, 2022). The EU Deforestation Regulation (EUDR) is a legislative measure in order to minimize the EU's contribution to deforestation and forest degradation worldwide. It requires operators and non-SME traders who place certain commodities on the EU market or export them from there to ensure that their supply chains are legal and deforestation-free. The EUDR is aligned with several international declarations and policy goals of the EU, the German Federal Ministry for Economic Cooperation and Development (BMZ), and partner countries in the (sub)tropics. The EUDR contributes to the implementation of various important global agreements such as the Sustainable Development Goals (SDGs), the Paris Agreement, the Kunming Montreal Global Biodiversity Framework and the Glasgow Leaders' Declaration on Forests and Land Use which emphasize the critical need to protect the world's forests. The EUDR entered into force in June 2023 and is planned to enter into application from 31 December 2025 onwards.

The EUDR requires operators and non-SME² traders placing commodities and products on the EU market or export from EU to exercise due diligence for all relevant products with the objective to ensure that products were produced legally and "deforestation-free". Due diligence for operators entails three elements: information, risk assessment and risk mitigation. Identifying the geo-location of the plot of land where the commodity was produced is one of the core elements of the EUDR's information requirements. Providing this information shall ensure strict traceability of the commodity along the value chain to the respective plot of land. Although submitting a due diligence statement is only mandatory for operators and traders and not Lao companies, these companies can gain a competitive advantage if they can supply their buyers with the required information such as geolocation.

In the Lao context, the most relevant commodities under the scope of EUDR are coffee, cattle, rubber, and timber. As an ex-ante assessment on the potential effects of the EUDR showed (F4 - Forest For the Future Facility, 2023), direct exports to EU markets form a small part of Lao agricultural products and timber products and only play a certain role for coffee, with about 13% of coffee export to the EU in 2021 (ITC, 2023). A large part of the commodities is exported to regional markets, e.g. to Vietnam, Thailand or China, especially rubber, cattle and timber and may be further exported by operators from these countries to the EU market. In this regard the EUDR brings significant implications for Lao producers, processors and exporters as well as operators and traders sourcing Lao products.

In response to these emerging requirements, this study examines the existing traceability systems and tools in Laos, particularly providing information on the state of the art of supply chain traceability systems and the capacity of Lao operators to track commodities and products from the plot of land where the commodities have been produced to the point of sales to an EU operator.

In this regard, this study assesses the current state of existing traceability systems in Lao companies, smallholders, cooperatives across the coffee, rubber, cattle and timber sectors and how these systems can support Lao supply chain actors to provide such information to their existing or potential business

² According to the EUDR legal text, the term 'operator' means any natural or legal person who, in the course of a commercial activity, places relevant products on the EU market or exports them from there. 'Trader' means any natural or legal person who, in the course of a commercial activity, places relevant products on the EU market or exports them from there.

partners. In detail, the objectives of the present study are to: (i) review existing traceability systems and tools available in the Lao context, (ii) review the lessons learned during the FLEGT VPA process and the development of a Timber Legality Assurance System (TLAS) which includes supply chain control (traceability) and apply these lessons to other commodities, (iii) synthesize the findings of the study to give recommendations for traceability systems for Lao companies, smallholders, cooperatives, and policy makers to provide accurate traceability information for operators. In addition, this study serves as a case study and provides more detailed information on traceability based on the general information provided on traceability by the global study on available traceability system across countries (GIZ, 2024). To gain a better understanding of possible challenges as well as opportunities for aligning private sector practices with the requirement of traceability, this study describes the traceability practices currently used by key sectors in the Lao PDR. This study is aimed at identifying effective strategies and practical tools to ensure traceability aligned with the EUDR requirements.

2. Methodology

The methodology employed a combination of desk view of relevant documentation and a survey with stakeholders from relevant commodities to gather data and information. The findings were analysed and reported, with the preliminary results presented together with another study on the national legislative framework in Laos during a debriefing session with the EU Delegation in Laos and representatives from concerned government departments.

2.1. Desk Review

To achieve the objectives of this study, a desk review was conducted to provide an overview on sectors and inputs for the survey design, and to collect and analyse existing information on traceability systems used in Laos. Key sources included the European Union Deforestation Regulation (EUDR), the Global Traceability study (GIZ, 2024), and the ex-ante assessment of the EUDR 2022 (F4 - Forest For the Future Facility, 2023). In addition, complementary documents such as relevant articles, grey literature, government statistics, production and trade data, lessons learned from the FLEGT-VPA Lao TLAS framework, and presentations from relevant commodity associations were examined. A full list of documents and references used in this review is available in Annex A.

For the commodity timber, only a desk review was implemented. The Lao TLAS developed by Lao stakeholders defines the timber supply chain control of timber producers and timber processors to ensure the traceability of timber to the forest site and procedures require timber producers to provide the address of the timber and volumes/weights along the supply chains to be able to validate and reconcile its correctness and authenticity (see Section 3.1). Although the desk review provides a general understanding of the timber traceability system, it may not capture all the nuances of real-world implementation and challenges faced by stakeholders on the ground.

2.2. Survey

To complement the desk review, a survey was designed and conducted to gather primary data from private sector stakeholders in the following three sectors: coffee, rubber, and cattle. The survey aimed at a better understanding of the agricultural practices of relevant stakeholders by means of case studies (see Chapter 4) and provided on-the-ground insights into the supply chain and traceability systems used in Lao PDR.

Survey design

A structured questionnaire was developed for the interviews of key stakeholders. It is divided into five sections: (1) General Information; (2) Supply Chains; (3) Traceability Systems; (4) Support and Future Needs; and (5) Additional Comments, offering stakeholders the opportunity to provide open feedback and suggestions. The selected interview setting is a semi-structured approach, which is channelled by an interview guideline, however, allowing deviation from the suggested structure. Open questions enabled the interviewees to answer in their own words rather than selecting from a predefined set of answers, while closed questions allowed for consistency and comparability of the gathered data. The findings of the preliminary literature research laid the basis for the interview questionnaires (see Annex D for the Questionnaire).

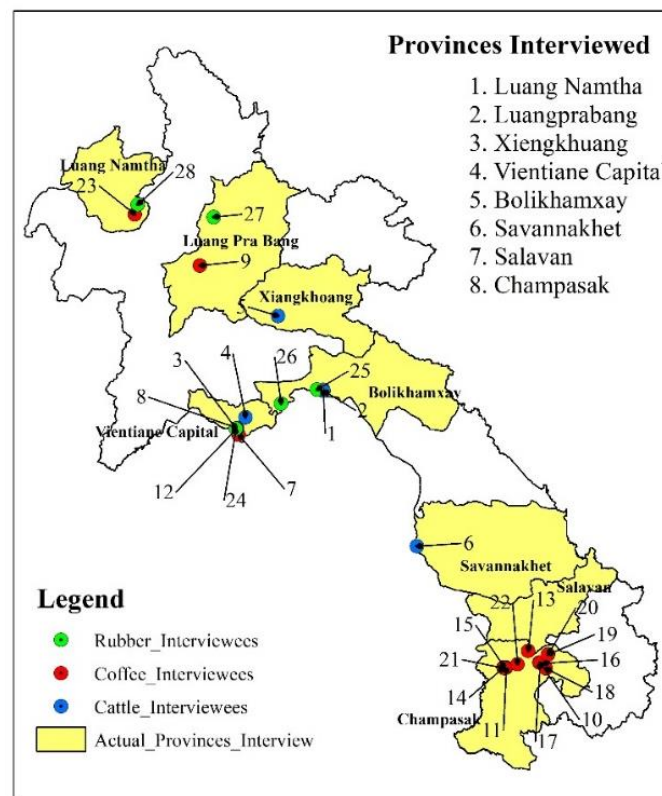
Sampling

The target respondents included companies, smallholders, associations, cooperatives from coffee, rubber, and cattle sectors in the Northern, Central, and Southern regions of Lao PDR and one international entity, supporting a project in the coffee sector. For a detailed description of the selection process of stakeholders and criteria details see Annex E.

Data Collection

The survey was conducted over three week (23 September 2024 to 14 October 2024) through on-site interviews at companies' premises or offices, the GIZ office in Vientiane Capital, telephone calls, and written questionnaires. Individual interviews have been conducted across the Northern, Central, and Southern regions of Lao PDR.

Figure 1: Map showing the geographic distribution of interviewees' activities



Source: Author's own illustration

The geographic distribution of the interviewees' activities, as shown in Figure 1, highlights the regions and provinces covered during the study. Interviewees were contacted via an official letter from Lao National Chamber of Commerce and Industry (LNCCI) in order to enhance the trust of private sector stakeholders into the interviewer and to provide the interviewer with official authority to conduct the interviews. The data collection process ensured anonymity, and confidentiality. Stakeholder profiles are provided in Annex B, where they are anonymized and numbered. These numbers are used for citation or reference in the report.

Data Analysis

Thematic analysis (Braun, 2012) was used to interpret qualitative data. This method involves comprehensive reading of each completed questionnaire to identify central themes and key terms that emerge across interviews by categorizing responses based on recurring keywords and patterns, the analysis offers a structured view of stakeholders' practices. Additionally, quantitative analysis was also applied, categorizing responses from multiple-choice questions to provide numerical overview of traceability adoption across stakeholders, and how their traceability is recorded. This dual approach combined qualitative insights with quantitative trends to enhance the comparability and detail of the findings.

Limitations

The study encountered several limitations that impacted the depth and comprehensiveness of its findings. A primary challenge was the limited availability and time of interviewees. Additionally, the limited cooperation of interviewees due to lack of long-term relationship, resulting in no responses, and cancelled interviews. Certain stakeholders have been interviewed but were unwilling to share their records and/or data forms with the interviewer. Furthermore, many interviewees were hesitant to share sensitive data or lacked sufficient insight into the subject matter, leading to incomplete or unclear responses.

Moreover, a significant number of interviewees demonstrated a limited understanding of the EUDR regulations, including its objectives and implementation targets. Many stakeholders were unfamiliar with the specific requirements and goals of the EUDR, which affected the quality of their responses. This lack of awareness about the EUDR's implications led to difficulties in obtaining precise information relevant to traceability and compliance expectations.

3. Overview on Sectors

3.1. Timber Sector

- Lao PDR is adapting its Timber Legality Assurance System (TLAS) after the termination of the VPA negotiations with the EU into a Lao TLAS to ensure timber legality and traceability.
- Management of the wood processing sector shifted to MAF, with DOF leading efforts on supply chain control, processing standards, and a national wood processing strategy.
- Key timber exports go to Vietnam, China, and Thailand, while EU markets are currently not a destination. Yet, operators and traders from main export destinations who produce finished products which do enter the EU market, would request Lao suppliers to provide information on legality and deforestation-free origin of products to be passed along the value chain under the EUDR.

- Timber supply chains are monitored at critical control points, with a transition planned from paper-based to cloud-based systems for enhanced traceability. Geolocation of the most important timber sources is legally required but currently not fully compatible with EUDR requirements.

Background

In 2017, Lao PDR started to design a Timber Legality Assurance System (TLAS) in the context of the negotiations between the Government and the European Commission on a Voluntary Partnership Agreement (VPA) and based on requirements defined under the FLEGT initiative. The TLAS has been described in the Draft Annex II (Lao-EU FLEGT VPA Negotiations, 2023). The factual termination of the VPA negotiations created the opportunity to refine the TLAS design and adapt the system into a Lao due diligence system for the forest and timber industries, by strengthening Lao objectives and circumstances. This adaption process to a Lao TLAS is currently ongoing under the lead of the Department of Forest Inspection (DOFI) and the Department of Forestry (DOF) under the Ministry of Agriculture and Forestry (MAF), based on a multi-stakeholder approach, and supported by the FLEGT-FC (KfW) and ProFEB (GIZ) projects. Legal verification and traceability of timber through the supply chain are still shaping the core elements of the Lao TLAS which is endorsed by the Forestry law 64/NA (2019), article 43 and the implementation is further specified by relevant legislation, partly still missing or to be amended. It is planned that the system will respond to specific requirements of international markets (e.g. EU/EUDR), although objectives and technical details are still being discussed among stakeholders. At any rate, the implementation of the Lao TLAS will allow operators better access to verifiable information that timber has been produced with the relevant legislation of Lao PDR as part of their due diligence obligation.

The longstanding FLEGT-VPA process and the current Lao TLAS development process make the timber sector most likely the best assessed of the four sectors included in the study, both in terms of legality and traceability of its products. Moreover, the awareness of sector stakeholders about legality and traceability issues is in general higher and will increase even more with the piloting of the Lao TLAS in three provinces (Luang Prabang, Vientiane Province, Savannakhet) which will start in early 2025 with support of FLEGT-FC, and later on the scaling up nation-wide.

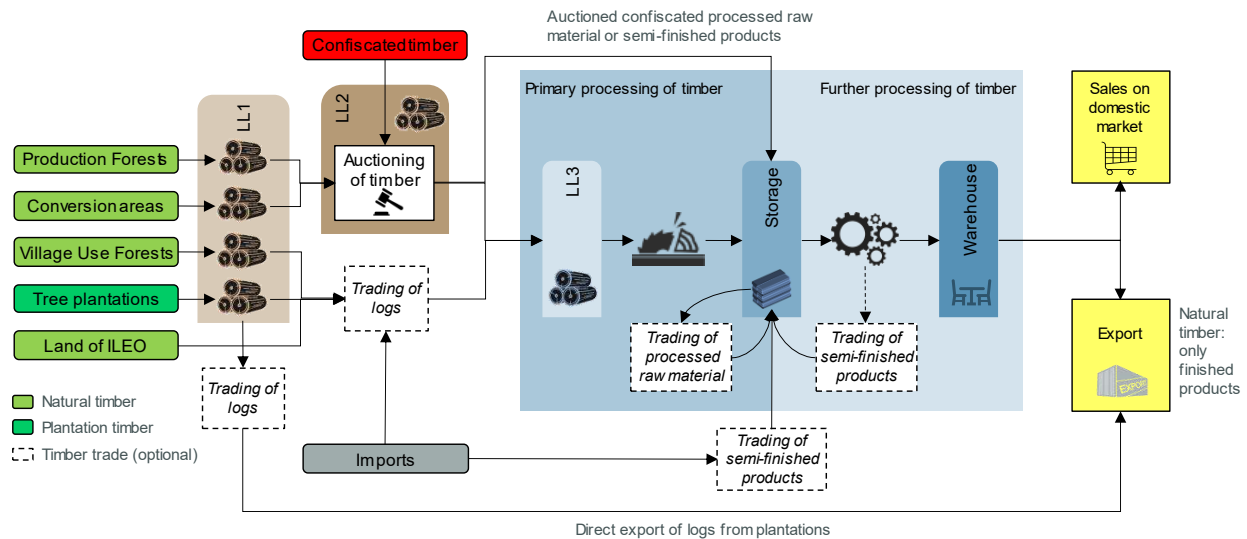
The ex-ante assessment (F4 - Forest For the Future Facility, 2023) mentioned that the responsibility to manage, promote and develop the wood processing sector was shifted in 2021 from MOIC to MAF. Export of wood products remains under MOIC/Department of Foreign Trade (DOFT) previously DIMEX). DOF has in the meantime started to integrate wood processing and trade related legislation in the legal framework under its responsibility and adopted the decisions on wood processing industry standards and on input-output monitoring. The newly established Timber and NTFPs Business Management Division under DOF plays a central role and is responsible for the development and dissemination of the legal framework on supply chain management and control for timber and wood processing and trade.

Supply Chains

Supply chains in the timber sector of Lao PDR have been intensely analysed by stakeholders in the FLEGT-VPA process, together with the legal requirements for supply chain management of timber sources and for wood processing and trade. The outcome has been described in the draft Annex II of the VPA (Lao-EU FLEGT VPA Negotiations, 2023). Figure 2 shows an overview on timber flows and supply chains for the seven identified timber sources (production forests, conversion areas, village use forests, tree plantations, land of individuals, legal entities or organizations (ILEO), confiscated timber and imports). Natural timber from production forests or conversion areas is stored at log landing 1 (LL1) close to harvest sites, further moved to LL2 where consolidated logs from different harvesting compartments are scaled

and graded, then auctioned and eventually transported to the log yards of primary processors (LL3). Confiscated timber must also be auctioned. Timber from village use forests, tree plantations, land of individuals, legal entities or organizations or imported logs are traded or sold directly to processors. Logs from plantations can directly be exported. Natural timber must be processed in-country and can only be exported as finished products.

Figure 2: Overview timber flows and supply chains



Source: (Lao-EU FLEGT VPA Negotiations, 2023)

Supply chain control is based on critical control points where materials from unregistered sources could enter or where registered materials could leave the system. The critical control points are defined by the forestry related legislation and ensure traceability and transparency of materials moved and this is being verified in the Lao TLAS by the verification body (DOFI/POFI) who validates data on products at critical control point and checks if data, e.g. volumes, is consistent between stages of the supply chain.

The main characteristics of the seven sources of timber, respectively how wood products are produced have been summarized in chapter 4.2 of the ex-ante assessment (F4 - Forest For the Future Facility, 2023). An analysis of the forestry and wood processing sector (UNIQUE Forestry and Land Use GmbH, 2021) estimated annual timber supply potential over the next five years to range between 1.1 and 1.9 million m³ roundwood and Timber plantations have been identified as the main source of roundwood representing about 90% of total domestic timber supply. The study concludes that it remains challenging to compile complete and consistent data on timber production and wood processing. Even information about smallholder plantations is incomplete due to a low rate of registration³ and information on natural trees on “private” land, owned by individuals, legal entities and organizations (ILEO), is not yet completely available, although timber resources on these lands may be substantial.

(F4 - Forest For the Future Facility, 2023) assessed trade data from the ITC trade map (ITC, 2023). ITC trade map data has been compiled from UN Comtrade⁴ until 2021. For the category “Wood and articles of wood, wood charcoal” (HS code 44, which includes fuelwood, wood in the rough and sawn wood, amongst others), the export values in 2010 and 2021 are nearly the same (EUR 30 million), while 2015 peaked to EUR 86 million, due largely to peaks in sawn wood and wood in rough exports. In 2021 the

³ FLEGT-FC recently estimated that around 5% of plantations are registered

⁴ [United Nations Comtrade database](#) aggregates detailed global annual and monthly trade statistics by product and trading partner

latter two categories decreased to < EUR 1 million. It can also be noted that in 2021, Vietnam was the largest importer of these wood products, followed by China, and Thailand far behind (see Table 1).

Table 1: Total wood production and exports by Laos and main destinations in 2010, 2015 and 2021

	2010			2015			2021		
	Volume (kt)	value (mio EUR)	% of total	Volume (kt)	value (mio EUR)	% of total	Volume (kt)	value (mio EUR)	% of total
Total exports (44*)	N/A	30.8	100%	N/A	86.2	100%	N/A	30.2	100%
Viet Nam	N/A	6.2	20.1%	N/A	54.0	62.6%	N/A	15.8	52.3%
China	N/A	2.2	7.1%	N/A	11.5	13.3%	N/A	7.2	23.8%
Thailand	N/A	21.9	71.1%	N/A	16.9	19.6%	N/A	2.7	8.9%
South Korea	N/A	0.2	0.6%	N/A	0.8	0.9%	N/A	1.7	5.6%
Total exports (4401**)	N/A	1.5	100%	N/A	0.5	100%	2.2	0.4	100%
Thailand	N/A	0.2	13.3%	N/A	0.063	12.6%	2.2	0.4	99.3%
Kuwait	N/A	0	0.0%	N/A	0	0.0%	N/A	0.0	0.70%
China	N/A	1.1	73.3%	N/A	0.116	23.2%	N/A	0.0	0.0%
Viet Nam	N/A	0.1	6.7%	N/A	0.306	61.2%	N/A	0.0	0.0%
Total exports (4403***)	N/A	1.1	100%	N/A	20.2	100%	16.4	1.1	100%
Viet Nam	N/A	0.4	39.8%	N/A	18.6	92.2%	15.3	0.9	84.2%
China	1.6	0.6	58.1%	N/A	0.8	4.1%	0.8	0.0	3.4%
Total exports (4407****)	35.7	15.8	100%	103.8	40.6	100%	6.5	0.9	100%
Viet Nam	3.2	2.4	15.4%	88.2	27.1	66.7%	1.4	0.6	70.7%
Thailand	32.5	13.3	84.5%	10.0	7.6	18.8%	0.8	0.2	21.4%
China	0.0	0.0	0.1%	4.3	5.8	14.2%	0.5	0.1	6.5%

(ITC, 2023)

* HS code 44 – Wood and articles of wood, wood charcoal

** HS code 4401 – Fuel wood

*** HS code 4403 – Wood in the rough

**** HS code 4407 – Sawn wood

No data for Lao PDR is available for 2022 and 2023 and it can be assumed that the conclusion drawn in the ex-ante assessment that EU27 is not a destination for the exported wood products is still valid and that operators and traders from main export destinations like Vietnam who produce finished products which do enter the EU27 market, would request Lao suppliers to provide information on legality and deforestation-free origin of products to be passed along the value chain under the EUDR. Lao wood processing companies who want to tap into new markets in the EU would also have an interest to provide information on deforestation-free/forest degradation-free and legal production.

As part of the of the ARISE Plus – Lao PDR project, funded by the EU (ITC, 2021), MOIC in collaboration with MAF and support and guidance provided by the International Trade Centre (ITC) developed a wood processing sector export roadmap. Initial discussions are currently held between government, ITC and stakeholders, for the preparation of a new project on trade and private sector development⁵ and if this roadmap should be updated under the new mandate of MAF. At the same time, DOF has been tasked by

⁵ Communicated by ITC representatives during project scoping meeting with ProFEB Forest Policy on 16/10/2024

the Ministry to draft a Wood Processing Strategy as under its Forest Strategy (Ministry of Agriculture and Forestry, 2024).

According to (ITC, 2021) and based on MOIC statistics, 1,325 wood processing establishments were operating in Lao PDR in 2016, dominated by small and medium-sized enterprises. By 2020, the number of wood processing factories had dropped to 1,030 manufacturing plants (8 sawmills, 383 wood processing factories and 657 furniture manufacturers) and 111 households-scale units. Yet, it is worth noting that processing facilities are often a combination of the aforementioned, e.g. establishments involved in both sawmilling and furniture production. MAF is currently updating the numbers with support from FLEGT-FC.

FSC certification can be a door-opener for EUDR readiness in terms of traceability requirements as the FSC standard defines specific traceability solutions. FSC is currently aligning its certification scheme with the EUDR focus on deforestation-free supply chains and introduces a regulatory module that requires a risk assessment and mitigation⁶. As a certification scheme, FSC will not constitute a green line to fulfil due diligence requirements under the EUDR, but it can help operators and traders with collecting information relevant under the EUDR (GIZ, 2024). Generally speaking, few plantation timber producers and wood processing companies have been certified in the past and only very few are still remaining keeping the door-opening function of FSC certification quite limited.

Discussions and preliminary work around a national forest certification system, endorsed by PEFC International, have been launched by DOF and a stakeholder consultation held with support from the UN-REDD Programme in 2020 (Sustainable Forest Trade in the Lower Mekong Region, 2020). Although progress has been slow, it gained again momentum, and a draft Lao National Forest Standard has been discussed and commented by a technical committee under lead of DOF in early December 2024. Promoters of a national standard expect advantages compared to international voluntary standards, such as national ownership, adaptation to national conditions or cost effectivity for smallholders, forest managers and the industry with the use of national auditors and accreditation body (PEFC, 2021). However, the capacity of auditors needs to be developed, and the existence of a national accreditation body is unclear. Moreover, the link to the Lao TLAS needs further discussion.

Traceability

With the endorsement of the TLAS in the legal framework of Lao PDR and subordinate regulations on supply chain control, timber products and operators have to comply with legal requirements and supply chain control becomes mandatory for all value chain actors. Yet, the legal requirements are not conclusively defined for all timber sources and important details for supply chain control at the wood processing industry and for trade are still missing (see section 5.1 for more details). This is the case for confiscated timber as one source of timber in Lao PDR which is not compatible with EUDR at all as confiscated timber can be traced back to the location of seizure but will only in exceptional cases be traceable back to the forest stand.

Current traceability systems for natural timber from Production Forest Areas and conversion areas and for plantation timber are almost exclusively paper-based. The systems document product specifications like felling and transport dates, dimensions, species, qualities and keep track of sales and as such fulfil important information requirements of EUDR as stipulated in article 9. The data structure of the forms in the paper trails along critical control points links the sequence of forms by unique identifiers which allow tracing back timber to the forest stand:

⁶ Video [Introducing FSC Aligned for EUDR](#)

- During harvesting in Production Forest Areas, logs are marked with paint by a unique code (forest address) after cross-cutting and can be traced back to the marked stump in the harvesting compartment (system of Reduced Impact Logging with single tree harvesting). The objective of this system is to allow location of harvested trees in the forest stand and check selective logging and not primarily the geolocation;
- For conversion areas, logs receive a unique code as well and can be traced back to harvest blocks (salvage logging system) on the harvest map (see Figure 3). Main objective is volume control of the operation (pre-harvest inventory data with harvesting data);
- Batches of plantation timber can be traced back to the plot via GPS coordinates recorded with Google Maps on a smart phone for the application of the plantation registration certificate. This is in accordance with geolocation digital formats accepted by the EUDR. Yet, the area threshold for points and polygons is different. This will be further discussed in section 5.1.

Figure 3: Coding of logs at LL2 in conversion area



Traceability of timber and timber products through the supply chain is one of the key objectives of the Lao TLAS and currently further developed in the adaptation process after the termination of the VPA negotiations. The draft Lao TLAS is designed as a cloud-based information management system (IMS). The IMS is planned as centralized data service (preferably cloud-based) with a modular architecture comprising of data entry with associated mobile apps, data storage, integrated data management between timber supply, processing, sales and compliance verification of the legality status of wood throughout the supply chain. Registered entities from Government, private sector and the general public will have access to the IMS as defined by specific user-rights. Timber is traceable to its origin and the mapping procedures and data structure may potentially include compatibility with EUDR digital geolocation formats, yet adoption of the existing procedures is not seen as a priority for the upcoming piloting. Timber sourced from producers from all seven sources will be registered and all processors and traders must introduce a supply chain control system. Activities, records and data are described for each critical control point in the supply chain and described in the annex to the legality definitions for timber sources, wood processing and trade and linked to the TLAS requirements. The piloting will start by applying the current paper-based system and registration of its data will be done manually in the IMS cloud and introduce step-by-step electronic systems by introducing handheld devices which will automatically upload the product data to the IMS.

3.2. Coffee Sector

- Coffee production is an important income source for many people in the Lao agricultural sector
- Lao coffee sector is dominated by smallholder production

- 13% of all Lao coffee exports are directly exported to the EU, other major markets are Vietnam and Thailand
- Application of comprehensive traceability systems is lacking, though certification schemes are largely used for export to EU

Background

Coffee plantations in Laos have been expanded over the past decades from 17,000 hectares in 1990 to 89,400 hectares in 2022 (FAO, 2022). More than 80% of coffee plantations are located in the southern part of Laos, particularly on the Bolaven Plateau (Champasack, Salavan, and Xekong Provinces), due to this area's mineral-rich soil. However, the Northern provinces of Laos – led by Luang Prabang – are increasingly gaining importance in coffee growing. The Lao coffee sector mainly produces Robusta and Arabica coffee beans, with some production of Liberica and Excelsa (ITC, 2021).

The Lao coffee sector is an important income source for many people in the agricultural sector of Laos, which accounts for 63.2% of the employment in 2018. The Lao Coffee Board estimates that the industry support the livelihoods of more than 100,000 people (ITC, 2021). Several studies indicate that coffee plantation can further create job opportunities and boost farmers' income, especially for those growing organic coffee and thereby accessing a higher-price market (JICA, 2012) (Seneduangdeth, 2018).

There are several plans and strategies in place to support the development of the Lao coffee sector. The Lao Coffee Sector Development Strategy by 2025, released in June 2014, is one of them and focusses on improving productivity and competitiveness (ITC, 2021). These goals shall be reached by increased value-creation within the Lao coffee sector, capacity building for and cooperation between relevant stakeholders. Other plans aiming at a more competitive and sustainable coffee sector in Laos are the Eighth and Ninth Five-Year National Socio-Economic Development Plan (NSDEP) 2016-2020, 2021-2025, the Agriculture Development Strategy to 2025 and Vision to 2025, and the SME Development Plan 2016–2020 (ibid.).

The Lao coffee sector is supported by the Lao Coffee Board (Conseil National du Café Lao, LCNC) and the Lao Coffee Association (LCA). The LCNC was officially created by the Lao Prime Minister in June 2010 and aims at the development of a strategic action to develop and promote the Lao coffee sector. Amongst others they also pilot, support and monitor associations, cooperatives and producer groups in the implementation of national policies and related laws (LCNC, 2021). The LCA consists of 67 private sector companies and 75 cooperatives, which are organized into six groups. These groups collaborate closely, sharing information and supporting each other around coffee industry operations. The LCA plays a crucial role as a service provider by acting as a centre for skills and knowledge enhancement, product distribution channels, and business support. Additionally, the association serves as an intermediary for government regulations and policies, helping to communicate them to the public and providing solutions for issues faced by its members.⁷

Recognizing the coffee value as contributors to the country's economic growth, the Lao government and development partners have been working together to enhance production, quality and market access for Lao coffee over the past decade (KPL, 2024). For instance, the Green CUP project, launched on March 14, 2024, is one of two flagship initiatives under the "Global Gateway Team Europe Partnership with the Lao PDR" as part of the TICAF program, aimed at enhancing sustainable and inclusive trade, investment, and connectivity in the agriculture and forestry sectors. The project focuses on strengthening trade and investment in the green economy, particularly within the coffee and tea industries, by promoting agro-ecological, community-based actions, climate-resilient value chains, and improved access to sustainable, high value markets, notably European market (Lapuekou, 2024).

⁷ Presentation from Lao Coffee Association 2024

Supply Chains

Coffee⁸ contributes to the economic development of the Lao PDR with an export value of 76 Mio EUR and 38,747 tonnes exported per year, making Lao coffee the 36th (value) and 27th world exporter (volume/weight). Vietnam and Thailand are the biggest export markets with an estimated export value of EUR 40,750 and 12,542 thousand in 2021. The EU-27 market represented 13% of the export value in 2021 while 54% went to Vietnam (see Table 2). From Vietnam, more than 45% of the total coffee exports with a value of EUR 828,766 thousand are going to the EU (see Table 3). From Thailand, more than 9% of coffee exports are going to the EU (see Table 4). If all Lao coffee imported by Vietnam or Thailand would be re-exported to the EU, Lao coffee would make up a share of 5% of Vietnamese and all of Thailand's exports.

Table 2: Lao coffee importers

Importers	Share in value in Lao PDR's coffee exports [%]			Value of Lao PDR's coffee exports, [EUR in thousand]		
	2019	2020	2021	2019	2020	2021
Vietnam	54	64.6	53.6	30,999	48,615	40,750
Thailand	15.6	8	16.5	8,954	6,010	12,542
EU-27	12	14.2	13.3	6,874	10,654	10,145

(ITC, 2023)

*Note: Data from 2022-2024 is not available

Table 3. Vietnam's coffee exports to EU.

Importers	Share in value in Vietnam's coffee exports [%]			Value of Vietnam's coffee exports [EUR thousand]		
	2019	2020	2021	2019	2020	2021
EU	47.4	47.8	45.5	938,663	827,219	828,766

(ITC, 2023)

Table 4. Thailand's coffee exports to EU.

Importers	Share in value in Thailand's coffee exports [%]			Value of Thailand's coffee exports [EUR thousand]		
	2019	2020	2021	2019	2020	2021
EU	6.5	9.6	9.3	236	252	288

(ITC, 2023)

The supply chain in the Lao coffee sector consists of many different stakeholders who each have a distinct role in harvesting, processing and trading the coffee. The largest share of coffee production derives from 20,000 to 25,000 smallholder farmers who own less than 3.5 ha of land and sell their coffee cherries to local collectors. The collectors then sell the product to other collectors and processors. However, there are also farmer organizations and cooperatives who collect the coffee cherries from farmers and afterwards process, sort, and grade them (F4 - Forest For the Future Facility, 2023). The Lao coffee cooperative units are particularly involved in supporting both smallholder farmers and large-scale company enterprise by facilitating strategies, market access, providing training in sustainable practices, and improving quality control by providing useful information to meet the production standard. Coffee cooperatives play a key role in empowering local farmers in Laos, not only promoting community development, gathering the products, exporting but also promoting Lao coffee into the international stage.

About one third of the total harvested coffee area are commercial plantations, who also largely take the role of the primary processor, before selling the produce to a secondary processor and/or exporter.

⁸ All numbers mentioned in this section refer to product Code 0901: Coffee, whether or not roasted or decaffeinated; coffee husks and skins; coffee substitutes containing coffee in any proportion

Approximately 10 companies are involved in the secondary processing and export of coffee, with the largest being responsible for about two thirds of total exports of Lao coffee. End market receivers do then take the responsibility for activities such as regrading, blending and roasting of coffee beans (ITC, 2021).

Traceability

For most coffee production, traceability back to individual producers is not established. Coffee is typically treated as a bulk commodity, with beans from various origins blended and grouped by factors like volume, price, and quality. Establishing traceability for coffee from commercial plantations that only process coffee from their own land is likely to be easier compared to smallholder plantations trading the beans to middlemen, who then mix it with beans from other farmers. For the latter case, providing details on the specific growing location is largely not possible. However, there are exceptions: Certified coffees and niche domestic markets, such as the Bolaven Plateau Coffee Producers Cooperative (CPC), may involve specific programs where a set group of farmers participates, allowing the coffee to be traced back to the farm. Additionally, organic coffee is kept separate throughout the supply chain, with a certain traceability maintained through detailed records and distinct handling, such as using separate bags, transportation, and storage. Details of traceability systems used by the producers are mostly not publicly available. Due to generic trade data which does not distinguish between coffee with and without certification, export volumes of certified coffee are unknown. However, the assumption is that all coffee exported to the EU is certified organic and/or fairtrade. Despite these efforts, geolocation technology is rarely used (F4 - Forest For the Future Facility, 2023).

Though literature on the challenges and benefits for installing a traceability system in the coffee supply chains of Lao producers is very limited, it can be assumed that incentives working for and against their instalment are similar to other sectors and regions. Firms decide to take up a traceability system to improve their risk management, have a better liability distribution among agents of the supply chain, and manage safety and quality requirements more effectively. Though traceability does not necessarily come with improved product quality itself, it often facilitates quality certification processes. Hence, the importance of certification schemes in the Lao coffee sector may work as an incentive for improving traceability (Stranieri, Cavaliere, & Banterle, 2016). Nevertheless, the dominance of small-scale coffee production, including mid-traders to transport the coffee from plantation or collection point to the processing factory as well as associated costs with increased traceability challenge this development.

3.3. Rubber Sector

- Area of rubber plantation has been growing rapidly in Laos over the last decades
- 30% of rubber production is managed by smallholders, 70% by foreign companies (concessions, contract farming)
- Vietnam and China are the major export markets for Lao rubber, however rubber might be re-exported to the EU
- Complex supply chains, lacking regulatory frameworks and insufficient law enforcement are key barriers for establishing traceability

Background

While rubber – *Hevea Brasiliensis* – was mainly grown in southern Thailand, south-eastern Vietnam and southern Myanmar in the mid of the 20th century, China's success in growing rubber in more "non-traditional" environments led to an increase of rubber plantations in other countries like Laos in the late

1990s (Fox & Castella, 2013).⁹ Within 30 years, the total area planted with rubber worldwide grew by 180%, adding up to 14.1 million ha in 2020 (Gitz, et al., 2020). While rubber plantation expansion had its peak between 2000 and 2011 in Laos, this development slowed down due to falling rubber prices and concession moratoria implemented in 2007 and 2012 (Traldi, et al., 2023). Nowadays, there are approximately 320,000 ha of rubber plantations in Laos¹⁰, including immature plantations that are not yet ready to be harvested for latex^{11,12}, as rubber trees take approximately 7 years to begin producing latex, reaching highest productivity only at around 20 years (ibid.). 56% of the rubber plantations are located in the north of Laos, the remaining plantations are in the central (19%) and southern (25%) parts.¹³

Depending on the quality of the latex as well as its processing, latex can be used for different products. Latex of lower quality can be used for vehicle tyres, car parts, food wear, while latex of higher quality is used to produce medical products such as gloves or condoms, food packaging etc (Haustermann & Knoke, 2019). Though natural rubber is competing with synthetic rubber made of crude oil, synthetic rubber cannot fully replace natural rubber due to its lower resilience, elasticity, abrasion and resistance. (Ahrends, et al., 2015).

The legal environment around rubber growing and production in Laos is still underdeveloped. There has been an agreement of MAF on rubber and processing management in October 2023 and MAF is developing a Prime Minister Decree on Natural Rubber Management.¹⁴ Different from Thailand or Indonesia, there is currently no rubber grade or national standard used for international trade in Laos, which could specify different grades for physical and chemical properties as well as cleanliness of the rubber. However, the government, together with partners from China is working on the development of such a standard (Smith, Lu, To, Mienmany, & Soukphaxay, 2020). Stakeholders in the Lao rubber sector, and in particular smallholders are supported by the Lao Rubber Association (LRA) which was established in 2019 and is currently in the process of attracting new members (Smith, Lu, To, Mienmany, & Soukphaxay, 2020).

Supply Chains

Major export markets for Lao rubber¹⁵ are Vietnam and China, which absorbed 64% and 36% of Lao exports in 2021 (Table 5). Considering natural rubber in primary forms, total exports from Laos (236,745 EUR thousand) are relatively small in comparison to Vietnam's exports of EUR 784,056 thousand but exceed China's exports of EUR 28,296 thousand by far. However, it has to be noted that Vietnam's and China's exports of processed rubber in the form of car tyres, tubes etc. exceed Laos' rubber exports of any kind by far. For instance, China exported EUR 2,612,786 thousand in new pneumatic tyres to the EU (ITC, 2023).

While Laos does not directly export its rubber to the EU, Vietnam and China export about 15.5% and 45% respectively of their natural rubber to the EU. If Vietnam would have re-exported all Lao rubber imports to the EU, this would have constituted around 96% of the Vietnamese exports of rubber in primary form to the EU in 2021. If China would have re-exported all Lao rubber imports to the EU, this would have constituted all of the Chinese exports of rubber in primary form to the EU in 2021 (Table 6, Table 7).

⁹ Presentation from Lao Rubber Association 2024

¹⁰ The reported data for rubber plantations varies significantly from source to source. For a detailed explanation see (Smith, Lu, To, Mienmany, & Soukphaxay, 2020).

¹¹

¹² Presentation from Lao Rubber Association 2024

¹³ Presentation from Lao Rubber Association 2024

¹⁴ Presentation from Lao Rubber Association 2024

¹⁵ All numbers mentioned in this section refer to product code 4001: Natural rubber, balata, gutta-percha, guayule, chicle and similar natural gums, in primary forms or in plates, sheets or strip

However, when looking at all rubber and articles thereof¹⁶, Lao exports to Vietnam and China would still make up a share of 30% and 2% of total exports to the EU in the respective countries.

Table 5. Lao rubber importers

Importers	Share in value in Lao PDR's rubber exports [%]			Value of Lao PDR's rubber exports [EUR in thousand]		
	2019	2020	2021	2019	2020	2021
Vietnam	55.1	49.4	64.4	107,086	92,788	152,475
China	44.4	50.6	35.6	86,342	95,210	84,270
EU-27	0	0	0	0	0	0

(ITC, 2023)

*Note: Data from 2022-2024 is not available

Table 6. Viet Nam's rubber exports to the EU.

Importers	Share in value in Vietnam's rubber exports [%]			Value of Vietnams rubber exports [EUR thousand]		
	2019	2020	2021	2019	2020	2021
EU	14.7	15	15.5	132,052	102,932	157,969

(ITC, 2023)

Table 7. China's rubber exports to the EU.

Importers	Share in value in China's rubber exports [%]			Value of China's rubber exports [EUR thousand]		
	2019	2020	2021	2019	2020	2021
EU	0.3	0.2	45.3	75	23	20,211

(ITC, 2023)

In Laos, 30% of rubber production are in the hands of smallholders, particularly in Central Laos where 50% of the rubber production is managed by smallholders. 70% of the production is in the hand of foreign companies, working on the basis of concessions and contract farming.¹⁷ The average size of a smallholder rubber farm is at 1.6 ha in Oudomxay and Luang Prabang province in Northern Laos, while the average size of company managed plantations varies significantly (Hicks, et al., 2009). In Laos, traders collect the coagulated¹⁸ latex from the farmers and sell it to further middlemen and processing factories.¹⁹ Cup lumps and rubber sheets can be further processed to block rubber and other forms of rubber sheets, to be processed into vehicle tyres, car parts, footwear and – if low contamination took place, medical products and food packaging. Stabilized latex on the other hand can be turned into premium vehicle tyres, food packaging, condoms and other high-value products (Haustermann & Knoke, 2019). In Laos, natural rubber is largely undergoing only primary processing – mostly into rubber blocks – before being exported to its main export markets Vietnam and China (Smith, Lu, To, Mienmany, & Soukphaxay, 2020). The export of raw,



Figure 4: Semi-processed rubber blocks

¹⁶ Product code 40: Rubber and articles thereof

¹⁷ Presentation from Lao Rubber Association 2024

¹⁸ "coagulated" refers to the process where the liquid latex transforms into a solid or semi-solid state. This happens when the rubber particles in the latex are made to clump together (coagulate), typically by adding acids like formic acid or acetic acid.

¹⁹ Presentation from Lao Rubber Association 2024

unprocessed latex is illegal in Laos according the MAF agreement on the rubber and processing management of 2023.

Traceability

Currently, there is neither a traceability system in place for rubber, nor is it legally required for export from Laos to Vietnam. In addition, awareness on international regulations requiring traceability such as EUDR is still very low among Lao stakeholders. Resultingly, companies do not pay special attention when it comes to mixing of rubber from different sources, such as smallholders who often supply bigger companies with rubber via middlemen (F4 - Forest For the Future Facility, 2023). Sometimes, latex from Laos, often in an unprocessed stadium as cup lumps or sheets, is exported to Vietnam and China and mixed with latex from other sources. The product is then labelled as Vietnamese or Chinese, creating problems for traceability. Furthermore, there are company reports about the theft of latex by farmers, who then sell the latex to independent traders (Smith, Lu, To, Mienmany, & Soukphaxay, 2020). Hence, the complexity of the value chain as well as the lack of a comprehensive regulatory framework for trading rubber can be considered key barriers when it comes to traceability.

Certification schemes, which could support traceability in the Lao rubber sector, are currently rarely used in the country. The Vietnamese Dak Lak Rubber Company, which also owns plantations in the Champassak Province of Laos, recently received its FSC certification (FSC FM/CoC) for 1,200 ha of rubber plantations in Vietnam and claims compliance of its rubber with the EUDR (FSC, 2024). The company is also a member of the Lao Rubber Association and aims at the FSC-FM certification of its rubber plantations in Laos.²⁰ Furthermore, the VietLao Rubber Company is certified with the PEFC certificate.²¹ Similar to the coffee sector, there is little research focussing specifically on traceability in the Lao rubber supply chain and incentives for its improvement. However, it can be assumed that drivers for the instalment of a traceability system are similar to those discussed in chapter 3.2.

3.4. Cattle Sector

- Size of cattle herds is continuously growing
- Cattle production predominantly relies on smallholders
- Live cattle are mostly exported to Vietnam, while beef is only exported to China
- Lack of animal registration and undocumented trading are key barriers for traceability

Background

The size of cattle herds was continuously growing in Laos since 2010 by an average of 5% per year, from 1.52 million in 2010/2011 to 1.88 million in 2016 (Napasirth & Napasirth, 2018) and 2.1 million in 2019 (Xayalath, Mujitaba, Ortega, & Rátky, 2021). The majority – 56% - of cattle are located in the Central region of Laos, 26% are located in the Northern and 17% in the Southern region. Despite the efforts of the Lao government to develop commercial-scale farms, 98% of cattle in Laos is still owned by smallholders, resulting in relatively low productivity of the sector. The most popular cattle breed in Laos is the Yellow Asian breed. Local cattle breeds reach up to 350 kg for male and 250 kg for female cattle. Since 2010, farmers imported different improved breeds such as the Thai Grey, Red Brahman and Thai Red Sindhi to improve meat production in order to meet domestic and international demand (Xayalath, Mujitaba, Ortega, & Rátky, 2021).

The Lao government aims at improving the productivity of cattle production as outlined in the “Agriculture Development Strategy to 2025 and vision to the year 2030” of MAF. It is aimed to increase cattle beef production from 14,000 tons by 2015 to 30,000 tons by 2020 and to 38,000 tons by 2025, by

²⁰ Presentation from DRI during rubber workshop 2024

²¹ Presentation from Lao Rubber Association 2024

allocating land for cattle production, changing the production system from traditional to commercial, develop linkages with processing factories and markets and dedicating land for the production of crops for animal feeds (F4 - Forest For the Future Facility, 2023).

Supply Chains

The cattle production in Laos predominantly relies on smallholders. Some farmers are also organized in cattle farmer groups and a few industrial farms have been established by investors (F4 - Forest For the Future Facility, 2023). Most farmers raise their cattle using traditional methods, such as free-ranging in fallow land and forests, free-ranging within fenced fallow, rotational grazing in fenced pasture, as well as the cut-and-carry system, where cattle are confined to pens and fed with harvested grasses or other vegetation (Phouyyavong, Tomita, & Yokoyama, 2019). While a large share of cattle is slaughtered within the country for domestic consumption, scholars have reported an increase of exporting live animals to China and Vietnam. Additionally, cattle are imported from Thailand for domestic use and Laos is used as a transit country for cattle trade from Thailand to Vietnam or China (F4 - Forest For the Future Facility, 2023). Live cattle trade to Vietnam largely takes place via the border in Xieng Khouang Province, while cattle is traded via Louang Namtha Province to reach China (Phonivsay, Ouanesamone, & Souvannavong, 2016).

While the export of live cattle²² is shifting from China to Vietnam (Table 8), the export of beef has already completely shifted from Vietnam to China (Table 9). There is no direct export from Laos to the EU (F4 - Forest For the Future Facility, 2023).

Table 8. Lao live bovine animals importers

Importers	Share in value in Lao PDR's live bovine animals ²³ exports [%]			Value of Lao PDR's live bovine animals exports [EUR in thousand]		
	2018	2019	2020	2018	2019	2020
Vietnam	88.3	96.7	98.8	67,505	195,861	216,428
China	11.7	3.2	1	8,954	6,555	2,280
EU-27	0	0	0	0	0	0

(ITC, 2023)

*Note: Data from 2022-2024 is not available

Table 9. Lao beef importers

	2015				2020			
	Volume (in tons)	Value (million EUR)	% of total	Unit value (EUR/ton)	Volume (in tons)	Value (million EUR)	% of total	Unit value (EUR/ton)
Total exports	27	0.2	100%	8,442	25	0.2	100%	913
China	0	0.0	0%	-	25	0.2	100%	913
Vietnam	27	0.2	100%	8,442	0	0.0	0%	-
Production (in thousand tons)	31.8 (with bones)				37.4 (with bones, 2020)			

(F4 - Forest For the Future Facility, 2023)

²² All numbers mentioned in this section refer to product Code 0102 Live bovine animals

²³ The term bovine animals includes cattle, buffalo and other bovine animals. Buffalo farms are by far less important than cattle farms. However, domestic water buffalos account for about a third of total heads of bovines from smallholders in Lao PDR. Yet, EUDR covers only cattle and not buffalo.

Over the past decade, the cattle population in Laos has also steadily increased, driven by an Export Agreement between Laos and China in 2017, which granted Laos a quota to export 500,000 cattle to China from 2021-2028. However, this figure has not been met due to the strict border restrictions imposed by Chinese authorities since the Covid-19 outbreak of 2020. Other agreements to increase livestock trading from Laos to China could not be implemented in large scale due to the incapability of Lao exporters to meet the Chinese requirements, such as low weight of cattle or the lack of testing facilities on the Lao side. Although the Lao and Chinese governments have signed multiple agreements since 2017 outlining detailed procedures for livestock trading, only two of the three concession companies authorized to construct quarantine facilities have received approval from the General Administration of Customs China (GACC) and the Department of Livestock and Fisheries (DLF) of Laos to quarantine and export cattle to China (F4 - Forest For the Future Facility, 2023). Long delays at border checkpoints have led to increased animal husbandry costs, and live cattle have lost weight due to longer holding periods. In addition, high cost for slaughtering in China which need to be borne by the Lao exporters complicate live cattle export. These challenges have prompted traders to shift their focus to the domestic market, where transactions can be made more efficiently and without the logistical complications associated with the export process. As a result, live cattle trading at border checkpoints has declined, with activity levels significantly reduced compared to pre-pandemic.

Traceability

Traceability in the Lao cattle sector is complicated by several reasons, such as the low rate of registration for animals. On average, only 3% of buffalos and 5% of cattle is registered. This lack of proper documentation hinders the monitoring of trade and compliance with health and hygiene standards needed for export (F4 - Forest For the Future Facility, 2023). Furthermore, a large share of cattle movement to Vietnam remains undocumented as cattle is moved to Nong Het district, which border Vietnam, and then traded via unofficial border points (Phonivsay, Ouanesamone, & Souvannavong, 2016). If documented, domestic and cross-border transactions are recorded using paper-based systems and no part of the SPS and traceability system is digitized (F4 - Forest For the Future Facility, 2023).

4. Case Studies

Chapter 4 provides a summary of the interviews and general findings from the case studies structured according to the sections of the research questionnaire.

4.1. General Information on Interviewees

In total, 28 interviews were conducted with representatives from the coffee, cattle and rubber sector. The majority of interviewees are part of the coffee sector and were largely located in the Southern part of Laos. 6 interviews in total were held with stakeholders from the cattle sector, mostly in Central Laos. For the rubber sector, 5 interviews in total – 2 in the Northern and 3 in Central Laos – were conducted (Table 10).

Table 10. Number of interviews per sector and region

Commodities	Number of interviewees			
	Total	North	Central	South
Coffee	17	2	4	11
Cattle	6	1	5	-
Rubber	5	2	3	-
Total	28	5	12	11

Table 11. Number of the operation type in main commodities.

Commodities	Type of Operation	Number of stakeholders
Coffee	Association	1
	Other	1
	Companies	11
	Cooperatives	4
	Total	17
Rubber	Companies	4
	Association	1
	Total	5
Cattle	Companies	6
	Total	6
TOTAL		28

The operations in the main products of coffee, rubber and cattle are distributed among different stakeholder groups. For coffee, there are 17 stakeholders: eleven companies, four cooperatives, one international entity categorised as “other” and one association. In the rubber sector, there are five stakeholders: four companies and one association. The cattle sector has six stakeholders, all of which are companies (see

Table 11).

Coffee Sector

Based on findings from the stakeholders’ interviews (Annex B), the coffee interviewees may be classified into three categories: Level 1 (small-scale stakeholder), Level 2 (medium-scale stakeholder), and Level 3 (large-scale stakeholder). These levels are defined by annual production capacity and each level shows similar characteristics, such as years of operations and market orientation (see Figure 5).

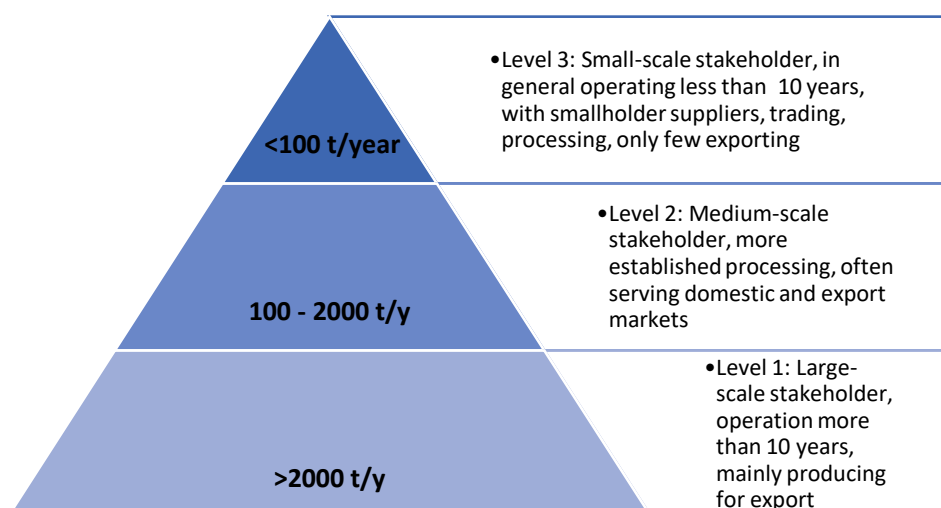
- **Level 1** stakeholders are large-scale producers with annual production exceeding 2000 tons and have been operating for more than 10 years (5 interviewees). They have great access to finance, and resources. Many of them are the key players in the industry, some of them are the founders and co-founder of Lao Coffee Association and contribute significantly to the country’s export revenues.²⁴
- **Level 2** includes the medium coffee producers or processors who have an annual coffee production between 100-2000 tons (5 interviewees). These stakeholders often have more established operations compared to level 1, allowing to serve both domestic and international market.²⁵
- **Level 3** covers the interviewees who produced or processed less than 100 tons of coffee annually and have generally been operating for less than 10 years (5 interviewees). Their primary business focus is on smallholders, trading, and processing. While most of these stakeholders’ supply to the domestic market, four interviewees have successfully entered international markets.²⁶

²⁴ Interviewed stakeholder 14, 15, 16, 21, 22.

²⁵ Interviewed stakeholder 13, 17, 18, 19, 20.

²⁶ Interviewed stakeholder 7, 8, 9, 10, 23.

Figure 5: Level of coffee interviewees



Source: Author's own illustration based on findings from stakeholder interviews.

Rubber Sector

Five interviewees from the rubber sector were interviewed, including representatives from four companies and one association. Most of these companies operate on land granted through government concessions, while some also purchase raw rubber directly from smallholder farmers. In terms of market orientation, all companies shared a similar approach, they export to neighbouring counties. Companies based in Central Laos primarily export their rubber to Vietnam,²⁷ while those in northern Laos mostly export to China.²⁸ The reason why there is no company exporting to the EU was because neither specific legislations nor rubber standards exist in Laos, moreover, they have limited capacity to process the raw rubber to products, and although their production volume is quite high, all the companies only exporting the raw latex. Three respondents reported annual production exceeding 20,000 tons per year, while one company mentioned that their output is less than 1000 tons per year.

Interviewee 24 highlighted the association's main activities, which include: enforcing association regulations to ensure compliance within the sector; promoting sustainable rubber production practices; supporting the operators in Laos in rubber production, processing and export; advocating the rights and interests of the operators and encouraging the small and medium enterprises (SMEs) to grow by providing capacity building and market access opportunities.

Cattle Sector

As shown in Table 10, there are six interviewees from the cattle sector, five of the respondents are located in Vientiane Capital, while one is based in Xiengkhouang Province. The sector is primarily dominated by smallholders who rely on traditional practices. All interviewees stated that their production is distributed within the domestic market, as they currently lack the capacity and resources to explore export opportunities. The interviewees highlighted several challenges, including limited knowledge of modern production processes,²⁹ insufficient access to advanced technology, and a lack of technical training programs, which constraints hinder their ability to enhance the productivity and meet higher production

²⁷ Interviewees 25, 26.

²⁸ Interviewees 27, 28.

²⁹ Interviewee 3.

standards³⁰. The production scale is relatively small compared to the other countries. Some interviewees reported producing fewer than 20 cattle per year, in contrast, two interviewees indicated annual production exceeding 100 cattle³¹. Additionally, the interviewees have been operating in producing cattle for a period ranging from 2 to 12 years.

4.2. Supply Chains

Coffee Sector

The interviews covered four cooperatives in the coffee sector³² who are all active in Champasack province. The members of these cooperatives are individual farmers who grow coffee on their individually owned plots of land, with one cooperative specifically promoting coffee in agroforestry systems³³. Interviewee 18 reports that the coffee cherries first undergo the process of wet and dry milling into parchment (white bean) at the village centre, until the cooperative collects the parchment for further processing into green beans. Interviewee 14 explains that the member villages of the cooperative are provided with a repulping machine to remove the outer skin and pulp of the coffee cherry from the centre. The village centre then serves as a collection point for the coffee beans before being sent to the cooperative's processing factory. However, some cooperatives³⁴ only undertake preliminary processing steps, such as drying the cherries, before selling to larger processors.

Local farmers do not necessarily have a formal contract with the cooperative that processes the coffee. Rather than that, the relationship is based on informal agreements where companies support farmers in sustainable practices, particularly shade-grown coffee under natural canopy cover, which enhances environmental resilience and coffee quality³⁵. Farmers also benefit from technical assistance provided by larger cooperatives, which helps to enhance both yield and quality.³⁶

Twelve processing factories, including one that is managed by a cooperative³⁷, were interviewed. Eleven companies³⁸ reported that they largely source their coffee beans from local farmers in addition to the – relatively smaller – harvest from their own plantations. Interviewee 9 reports that their own production amounts to approximately 20 tons per years, while 50-60 tons per year comes from around 800 local families. One company³⁹ reported to only use coffee from their own plantations in 2024, despite having bought coffee beans from local farmers in the past, due to the high prices they had to pay for the farmer's coffee beans, resulting into lower profitability of the company. While some respondents⁴⁰ state to not have a formal contract with their suppliers, many describe how they have built a long-term relationship with farmers through providing financial and technical support⁴¹, building coffee tree nurseries in the villages⁴², and offering trainings on coffee growing and processing⁴³ in addition to or as a substitute for formal agreements.

³⁰ Interviewee 4.

³¹ Interviewee 2, 4.

³² Interviewee 13, 14, 18, 19.

³³ Interviewee 19.

³⁴ Interviewee 19.

³⁵ Interview 19.

³⁶ Interview 13, 14.

³⁷ Interviewee 14.

³⁸ Interviewee 7, 8, 9, 10, 14, 15, 17, 20, 21, 22, 23.

³⁹ Interviewee 16 (used farmers in the past but not planned for 2024).

⁴⁰ Interviewee 10.

⁴¹ Interviewee 10.

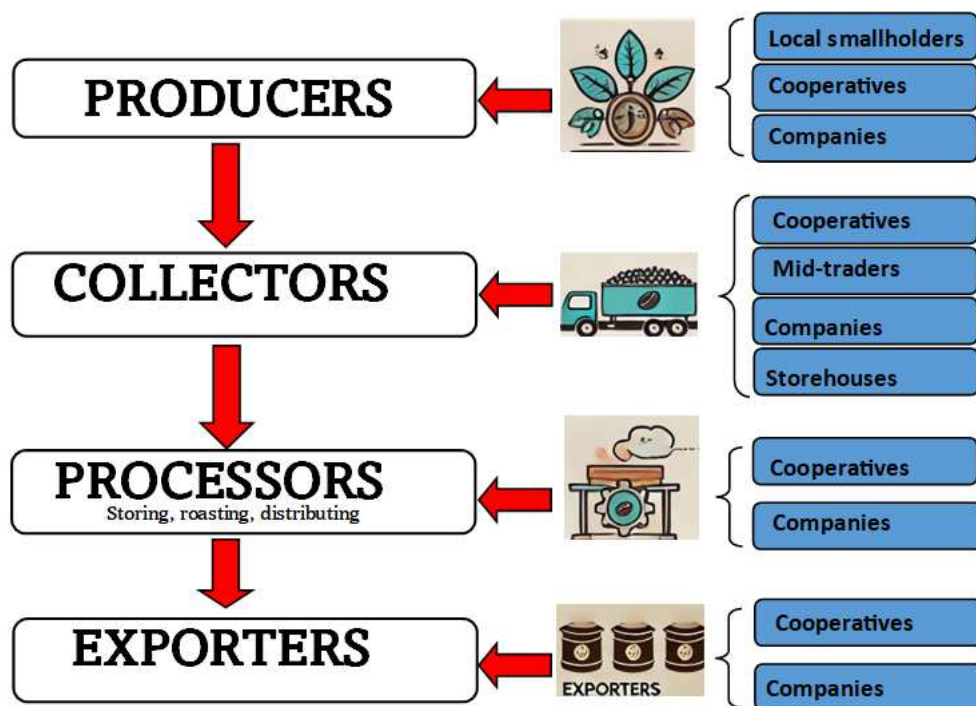
⁴² Interviewee 9.

⁴³ Interviewee 23.

Depending on the processing stage of the supplied coffee beans, the processing factories then wet wash and dry mill the coffee to produce parchment coffee. The parchment is then sorted and graded into different qualities before being milled into green beans. Approximately half of the respondents⁴⁴ reported to roast the coffee in their own factories, while one company⁴⁵ reported to further trade the green beans to a domestic roaster. The other half of processing factories report to export the beans for roasting in – mainly – neighbouring countries such as Vietnam and Thailand⁴⁶, while one company⁴⁷ exports to the EU for roasting.

Both, roasted and green beans are largely exported abroad, while only two producers report to sell their product domestically⁴⁸. Seven producers report to export their coffee to countries within the EU⁴⁹ – some with processing steps in neighbouring countries – and two companies report that they plan to export to the EU market in the future⁵⁰. Other markets are neighbouring countries such as Thailand, Vietnam, Japan, New Zealand and China as well as the United States⁵¹. The following Figure 6 depicts schematically the supply chains observed at the interviewed stakeholders.

Figure 6: Supply chains of visited coffee sector stakeholders



Source: Author's own illustration.

Rubber Sector

The interviews covered three companies – two Chinese and one Lao investors –, that grow their own rubber trees on concession land located in Northern and Central Laos⁵². Out of a total plantation area of

⁴⁴ Interviewee 7, 8, 9, 10, 16, 18, 21, 22.

⁴⁵ Interviewee 23.

⁴⁶ Interviewee 15, 16, 21, 22.

⁴⁷ Interviewee 17.

⁴⁸ Interviewee 16, 23.

⁴⁹ Interviewee 7, 8, 10, 15, 17, 20, 21.

⁵⁰ Interviewee 9, 23.

⁵¹ Interviewee 8, 9, 10, 16, 18, 20, 21, 22.

⁵² Interviewee 25, 27, 28.

320,000 ha⁵³, the concession area of one interviewee had a size of 14,000 ha.⁵⁴ One company directly sells its rubber collected from self-owned plantations to a Vietnamese semi-processing company located in Bolikhamxay. The other two companies also source rubber from smallholders living in the surrounding areas of the company in addition to producing their own rubber⁵⁵. In addition to the three companies that have their own plantations, one company – a Vietnamese investment – only sources from smallholders and does not have its own plantation⁵⁶. The smallholders providing the rubber in the form of cup lumps to the buyers often use traditional practices for growing and tapping the rubber and rarely receive any professional training on sustainable tapping techniques⁵⁷.

The rubber is purchased by middlemen who buy the rubber at village collection places before transporting it to different companies' processing plants (see Figure 7). Interviewee 28 states to have a formal contract with the mid-traders who supply the factory with the unprocessed rubber. Company number 26 sources its rubber from three different provinces but does not have a formal contract with suppliers. The rubber is transferred to the production sites for further processing, during which the latex is cleaned, dried and semi-processed into products such as rubber blocks. Company 26 reports to produce rubber blocks in Standard Vietnamese Rubber (SVR) 10 quality, which has higher dirt and ash contamination levels compared to SVR 3L or SVR5 and is used for low-cost products such as footwear, where purity is not the highest priority.

Figure 7: Village collection point for rubber cup lumps



After a Vietnamese buyer has picked up the blocks from company 26 for further processing in Vietnam, the rubber product is sold to Chinese buyers. Similarly, interviewee 28 reports to also export the processed rubber to China. None of the interviewed stakeholders reports to export to the EU or having any knowledge about rubber goods being sold to the EU after processing in neighbouring countries. Figure 8 shows a synopsis of the supply chains described by the interviewed stakeholders.

⁵³ Interviewee 24.

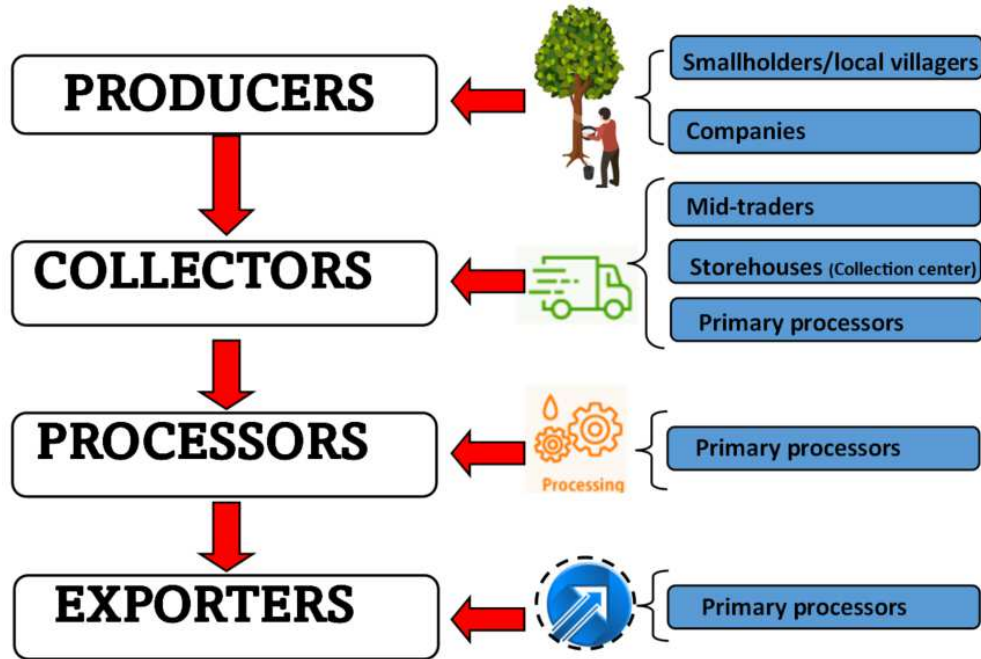
⁵⁴ Information not available from 3rd interviewee.

⁵⁵ Interviewee 27, 28.

⁵⁶ Interviewee 26.

⁵⁷ Information from field visit during the regional workshop on “Promoting Sustainable Natural Rubber Production, Trade and Investment in Mainland and Southeast Asia”, 15. November 2024, Vientiane Capital.

Figure 8: Supply chains of visited rubber sector stakeholders



Source: Author's own illustration.

Cattle Sector

The interviews covered six producers who are active in Northern⁵⁸ and Central⁵⁹ Laos. All interviewed stakeholder report to breed their own cattle at their own farm. Production is relatively low with three farms reporting to produce approximately 10-20 cows annually and only one⁶⁰ reporting to produce 100-150 of beef cattle on average⁶¹. The remaining interviewees either considered this information as sensitive and preferred not to report or did not have any knowledge about their annual production capacity. In addition to producing their own cattle, two interviewees report to buy additional animals from local farmers⁶². The interviewed producers focus on raising cattle, providing proper breeding, feeding and overall health management until the cattle are ready for sale.

After production, primary middlemen act as key intermediaries. These small-scale traders purchase the live cattle directly from farms and local smallholders, handle transportation logistics and deliver the cattle to larger market centres or other intermediaries for further aggregation⁶³. The process continues with secondary middlemen, who operate within major markets, such as the Xieng Khouang Livestock Trading Centre where people can exchange, buy and sell their products. Markets serve as key trading hubs, connecting traders, exporters and buyers. Livestock sold here may remain on domestic markets or be transported back to border checkpoints for international buyers. Exporters, typically large traders or companies, manage the logistics of cross-border trade, particularly to major markets such as China and Vietnam. Some exporters work directly at border checkpoints, while others rely on central markets for sourcing and documentation. In contrast to the majority of respondents who only sell live cattle, two

⁵⁸ Interviewee 5.

⁵⁹ Interviewee 1, 2, 3, 4, 6.

⁶⁰ Interviewee 4.

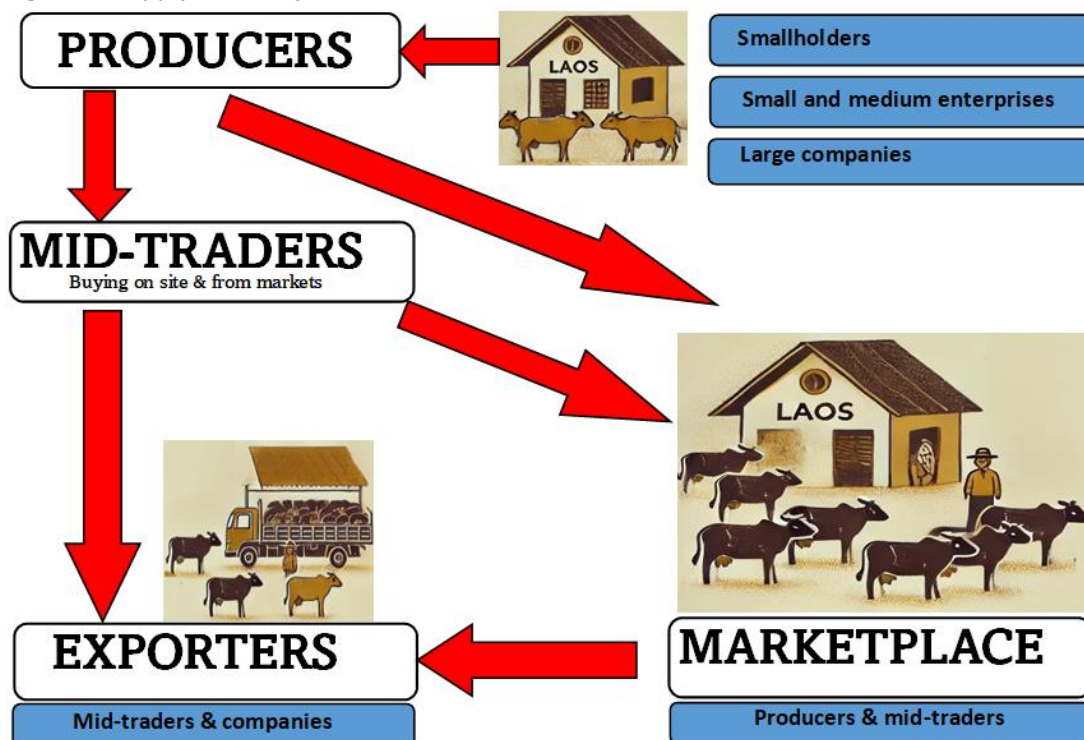
⁶¹ Interviewee 1, 5, 6.

⁶² Interviewee 3, 5.

⁶³ Interviewee 1, 2, 3, 5, 6.

interviewees also report to additionally⁶⁴ or only⁶⁵ sell meat to domestic markets such as Vientiane. Interviewee 4 owns its own slaughterhouse which processes about 15 cows per day and is certified according to a GMP-ISO standard. It is important to note that all interviewees report to only sell their products domestically and do not have any knowledge about possible exports to neighbouring countries. Supply chains reported by interviewees are visualized in Figure 9.

Figure 9: Supply chains of visited cattle sector stakeholders



Source: Author's own illustration.

4.3. Traceability

The results of the stakeholder interviews revealed several insights into the traceability practices across the three selected commodities in Laos. In the coffee sector, fourteen respondents reported using traceability methods, while one reported not using them. For rubber sector, 4 respondents reported using traceability methods. However, in the cattle sector, only half of the respondents (3) reported using traceability methods, while the other half indicated the absence of such practices (Table 12).

Table 12. Use of traceability systems among visited stakeholders.

Type of Commodities	Yes	No	Not applicable
Coffee	14	1	2
Rubber	4	0	1
Cattle	3	3	0
TOTAL	21	4	3

⁶⁴ Interviewee 3.

⁶⁵ Interviewee 4.

Table 13 provides an overview of the traceability systems used among different commodity groups (coffee, rubber and cattle) in Laos, highlighting the predominance of traditional paper-based methods. In the coffee sector, seven respondents report using paper-based traceability systems, followed by rubber with four and cattle with three respondents indicating their use. Additionally, five respondents from the coffee sector explain to document their trading data in a mix of paper-based and digital system indicating a gradual transition but highlighting limited integration of modern technologies. Solely digital traceability systems are rarely used, with only two respondents from the coffee sector implementing such a system, while no respondent from the rubber or cattle sector report to have a digital system, indicating a significant adoption gap due to barriers such as limited knowledge and infrastructure. A detailed description of traceability systems in use can be found in chapter 5.

Table 13. Type of traceability systems used by interviewees.

Commodity	Traceability system type		
	Paper-based	Digital	Paper-based and digital
Coffee	7	2	5
Rubber	4	0	0
Cattle	3	0	0

Though product certification does not necessarily improve traceability, it was identified as a significant factor for supporting traceability efforts of interviewees. As certification schemes often come with stringent requirements for documentation and transparency throughout the supply chain, producers using certification schemes such as organic or fair trade often had more comprehensive and sometimes digital systems in place in comparison to producers without certification.⁶⁶ This was particularly the case for producers in the coffee sector. Among the coffee respondents, eight reported holding some kind of certification, with the most common being Fair Trade, Organic, and Rainforest Alliance. A few respondents also held GMP (Good Manufacturing Practices) and other certifications, which enhance the traceability and facilitate market access to regions with stringent requirements, such as the EU. However, nine coffee respondents indicated that they lacked certification, stating that high membership fees significantly increase the cost of their coffee as the primary barrier.

In the rubber sector, only one out of the five respondents held a certification, specifically SCR (Standard China Rubber), which aligns with market demand in the region. In contrast, none of the respondents in the cattle sector reported holding any certification. Many expressed limited awareness of available certification schemes or cited the lack of practical incentives to pursue them.

4.4. Support Needed for Future Improvement

The survey respondents expressed various needs and required assistance in the field of capacity building, financial support, market expansion, equipment, knowledge on relevant rules and regulations, and infrastructure, that need to be addressed in order to enhance EUDR readiness in the Lao agriculture sector.

Starting with the implementation of a traceability system, a majority (59%) of respondents did not prioritize the need for support in establishing such a system, while 41% recognized its importance. This suggests that while traceability is important for some, many respondents may not view it as an immediate concern or are not fully aware of its potential benefits for their operations. Upstream actors do not see a need for a traceability system as they perceive it unnecessary, with no prior demand from

⁶⁶ Interviewee 14, 17, 18, 20

buyers or traders. Lacking awareness of its purpose, benefits, or implementation, their focus remains solely on producing and selling goods, without understanding traceability's role in market access, quality assurance, or compliance.

Capacity building emerged as a significant need and focused on improving digital literacy and creating awareness on the benefits of traceability systems. Training is needed in data collection and storage, usage of geospatial tools and compliance process with 74% of respondents expressing the necessity for training and development. This overwhelming majority signals a strong desire to enhance skills and knowledge within the workforce, indicating that respondents see capacity building as a crucial step for improving productivity and operational efficiency.

Similarly, investment and financial support is equally prioritized, with 74% of respondents indicating a need for financial assistance for both, collecting geolocation and other data as well as establishing and maintaining traceability systems. This reflects the respondents' recognition that funding is essential for sustaining operations, supporting growth, and addressing financial challenges faced in their respective sectors.

Targeted support for market expansion and compliance with international law and standards, such as EUDR requirements also saw substantial demand, with 78% of respondents emphasizing the need for support in these fields. Respondents' demand for support reflects a growing awareness of these regulations, emphasizing the need for robust systems to meet EUDR criteria and maintain competitiveness in international markets. This highlights a clear intention to engage with international markets, suggesting that respondents recognize the importance of aligning with global expectations to remain competitive.

On the other hand, technical and operational equipment needs were more evenly split, with 52% of respondents indicating a need for additional equipment, while 48% did not. This balance suggests that while some recognize the importance of modernizing equipment to enhance operations by using modern equipment like GPS devices, drones, and digital reporting tools to improve data accuracy, geolocation tracking, real-time monitoring, and efficient documentation of supply chain activities, others may already be equipped or prioritize different areas of development.

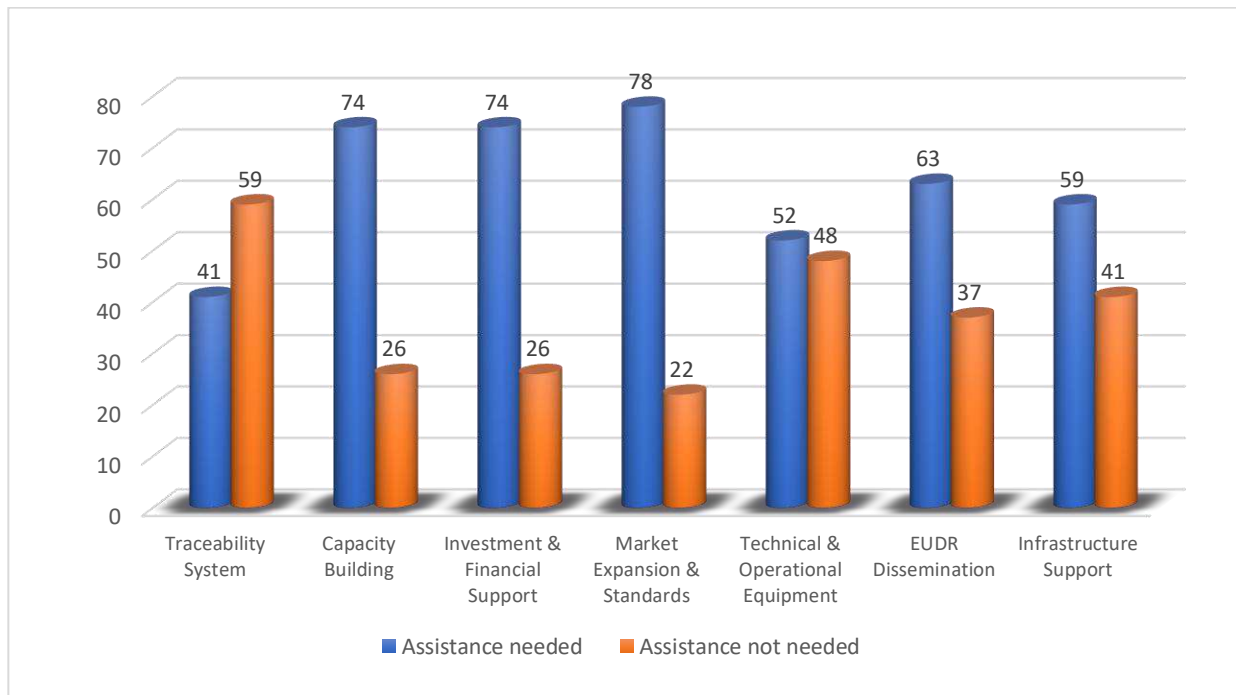
The survey also revealed a need for greater EUDR dissemination, with 63% of respondents calling for more awareness and understanding of the EU's due diligence requirements. This need was particularly expressed by stakeholders from the coffee sector, who are already partially exporting to the EU or plan to do so in the future. This underlines the necessity for clearer communication regarding regulatory requirements and a continuation of EUDR related outreach to stakeholder groups in Lao commodity sectors to reach a larger number of the interviewed stakeholders

Finally, infrastructure support was requested by 59% of respondents, highlighting the need for improved physical and logistical frameworks to better support operations. This could be crucial in addressing challenges related to transportation, communication, or distribution networks.

In summary, the survey findings reveal a clear prioritization of capacity building, financial support, market expansion, and infrastructure improvement. While the need for traceability and technical equipment is less pronounced, the demand for dissemination of information on the EUDR and financial assistance remains critical. These insights provide a foundation for understanding the key areas where stakeholders may need to focus their efforts to foster growth and competitiveness in their respective sectors.

The bar chart below (Figure 10) visualizes the responses for each of the seven categories, showing the percentage of respondents who indicated a need versus those who did not.

Figure 10: Support and Future Needs for different categories (responses in %)



Source: Author's own illustration based on findings from stakeholder interviews.

5. Review of Existing Traceability Schemes and Tools

5.1. Current Traceability Practices

In Laos, traceability systems for sectors like timber, coffee, rubber, and cattle from producers to traders, processors and exporters are at varying stages of development influenced by factors such as regulatory frameworks, common practices, regional and international market demand and requirements. The study shows that the commodities' traceability is paper-based and in some cases mixed with both paper-based and digital records. The sub-sections below briefly describe the traceability systems used in each sector with examples of practical traceability.

Timber traceability systems

The current implementation of traceability systems for timber and timber products is weak and stakeholders agree that the implementation of the TLAS will take considerable time beyond the expected application of the EUDR (F4 - Forest For the Future Facility, 2023). As part of the VPA process, verification procedures for supply chain control have been piloted in two exercises in 2020/21 and in 2022 (Schwab, 2022) at plantation companies, in a conversion area and for confiscated timber. The piloting allowed to draw conclusions on implementation status of traceability systems. Observed systems were paper-based and included pre-harvest inventory data, log lists, sales and export documents with exception of an FSC-certified plantation company with downstream processing which applied electronic systems for supply chain control. The piloting showed that the verification of paper-based systems is very time-consuming, and documents may not be readily available. This holds especially true for documents not directly related to product data in the supply chain but demonstrating legality of products and operations, like harvest permits, transport permits, environmental impact assessments of factories or documents demonstrating the fulfilment of labour obligations of companies. Supply chain control systems were only partly

established, and most plantation timber companies could not trace back timber from their smallholder suppliers as the plantation registration certificate was not available.

The review of the existing traceability system was an inherent part of the VPA process and started with the in-depth analysis of the legal framework by the VPA stakeholders during the establishment of the Timber Legality Definitions (TLD). The analysis allowed to identify inconsistencies in the legislation of a sector, or in-between different sectors involved in TLAS implementation, the key ministries being MAF, MOIC, MONRE, MOF and MOLSW. The multistakeholder process also addressed gaps regarding a sound supply chain control. As a result, responsible government entities started amending existing regulations or addressing existing gaps. With the redesign of the Lao TLAS, a reassessment of legal requirements regarding supply chain control and their compliance with the Lao TLAS requirement has been undertaken recently under the FLEGT-FC⁶⁷ and the annexes to the legality definitions been revised.

The following challenges regarding traceability and necessary action have been identified in the recent reassessment (Table 14):

Table 14: Remaining challenges and necessary action to ensure traceability of timber in the Lao TLAS

Supply Chain Type	Main Challenges	Necessary Action
Supply chain control in PFA and conversion areas	Harvesting operations are implemented by subcontractors, yet their roles and responsibilities are not clearly defined in the regulation on supply chain control (Instruction 0131/MAF (2021) On the management of logs movement from production forest areas, conversion areas and infrastructure development areas from log landing I to Log Landing II and Log Landing III.) and the existing Decision 0182/MAF (2008) On harvesting contractors is outdated Logging management for conversion area is regulated under a legislation from 2008 (Decision 0112/MAF (2008)) which was defining the management of salvage logging for hydropower projects. Other infrastructure development projects are not sufficiently reflected.	- review of 0131/MAF (2021) - amendment of Decision 0182/MAF (2008) - amendment of Decision 0112/MAF 2018
Village use forest	Progress have been made recently with respect to village forest management planning with the issuance of Instruction 4143/MAF (2023). Yet, supply chain control of timber from village use forest is not yet clearly defined, especially if responsibilities for harvest permits, issuance of transport permits and sales of timber can be delegated to the local level.	- Clarification of supply chain control in the village forestry legal framework.
ILEO	Supply chain control of timber from land of ILEO is addressed in Decision 3684/MAF. This includes registration of trees based on land use certificate, harvest permits and transport	- DOF acknowledged the weakness and started amending Decision 3684/MAF (2023)

⁶⁷ Since September 2024, FLEGT-FC organized 4 internal technical workshops with participation of DOF, DOFI and ProFEB and 2 TWG meetings with stakeholders from line agencies, private sector and CSO were held.

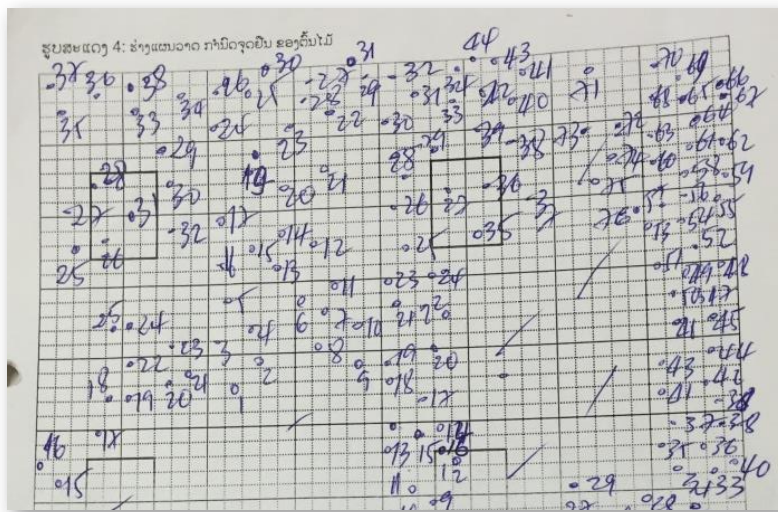
Supply Chain Type	Main Challenges	Necessary Action
	but does not clarify sufficiently supply chain control after felling and sales procedures.	
Supplies from smallholder plantations	The requirement to register tree plantations is going back to regulations issued in the early 2000. Yet, the implementation is still very weak. According to a new estimate from FLEGT-FC, only 5% of smallholder plantations are registered. The actual regulation, Instruction 2492/MAF (2020) regulates registration and reporting duties introduced in the Law 64/NA Forestry Law) before felling and before transport. During the registration process, plantation owners have to map their plantations by using Google Maps on a smart phone and indicate GPS coordinates on the application form.	- FLEGT FC will test the application of 2492/MAF (2020) in the upcoming piloting and recommend amendments if deemed necessary.
Confiscated timber	Supply chain control of confiscated timber is currently not covering all critical control points and traceability is not ensured.	- DOFI is currently drafting a Decision for management of confiscated timber
Imported timber	Timber is a controlled good and needs an import permit. Issuance of this import permit and Customs clearance is currently not linked to any due diligence. Under the VPA process, MOIC started to revise Decision 0451/MOIC (2012) On the Procedure for Timber Import to include due diligence requirements aligned with EUTR, yet the amendment has been deferred due to the unclear situation of the VPA process. Another challenge is that permits are contract-based, customs clearance is consignment based which does not allow traceability.	- An initial discussion has been held between FLEGT-FC and MOIC/DOFT how align the amendment of 0451/MOIC (2012) with the due diligence requirements of the Lao TLAS. - Customs shall be involved and the integration of timber imports and exports in the Lao National Single Window discussed.
Supply chain control system of wood processors and traders	Decision 1045/MAF (2023) On the Management and Following Up of Timber Input and Output does not include all requirements of the Lao TLAS to ensure traceability, e.g. daily registration of all material received, input into production batch units, associated product output and sales / exports.	- It is planned to develop an annex to Decision 1045/MAF (2023) which will further substantiate the requirements of supply chain control systems to comply with the Lao TLAS
Timber exports	Data structure of export documents from MOIC and for customs procedures from MOF are currently not consistent. Customs documents are missing relevant information as scientific names, HS codes.	- Inconsistencies have been identified and are being addressed in legal amendments - MOIC suggests to streamline export procedures and customs clearance by adding timber products to the Lao National Single Window.

As outlined in section 3.1, the traceability to the location of origin of source is part of the supply chain control in Production Forest Areas, conversion areas and for plantations but the information is only

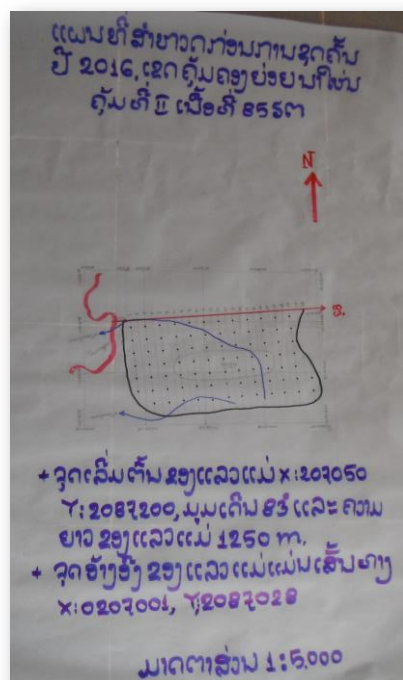
partially aligned with EUDR requirements with respect to geolocation in the case of plantations (with some caveats).

- PFA: location of trees is marked on paper-based pre-harvesting maps (recorded during the pre-harvest inventory) with a scale of 1:1000. The basis is a map of the harvesting compartment with a scale 1:5,000 as part of logging plan (see Figure 11). This map allows to locate a stump after felling (e.g. for mandatory post-harvest monitoring); yet the map is not easily legible and a digitization of the location of trees during pre-harvest inventory would improve the applicability of the system. This could be easily aligned with EUDR geolocation requirements; yet with the still existing logging moratorium, there is little priority to adapt the system.

Figure 11: Excerpt from a PFA pre-harvest inventory map and sample map of harvesting compartment



Excerpt from the sampled pre-harvest inventory map on a scale of 1:1,000. Zones with high density of inventoried trees are hardly legible
(Source: (UNIQUE Forestry and Land Use GmbH, 2016))



Sample map 1:5,000 of a harvesting compartment from a timber harvesting plan for 2016-2017 (Source: SUFORD-AF, 2016)

- Conversion areas: a map with the delineation of the area to be converted into harvesting blocks is required by Decision 0112/MAF 2008.

- Plantation timber: Instruction 2492/MAF (2020)) requires the applicant for a plantation registration certificate to record GPS coordinates of the plantation. A graph explains the procedure (Figure 12). Moreover, the instruction specifies whether a single point or polygons have to be recorded depending on the type of plantation owner and size of the plantation (Table 15). The instruction defines three types of plantation owners (English translation): a) individuals (mostly smallholders, normally referred to as households in the Lao context); b) organizations⁶⁸, c) entities (“legal entities” in the Lao version). The instruction requires:
 - o Individuals to register a single GPS point, independently from the size of the plantation
 - o Organizations to register a polygon OR a singly GPS point, independently from the size of the plantation
 - o Legal entities to register a polygon, independently from the size of the plantation

As the EUDR geolocation threshold for point/polygon recording is 4 ha, independently from the owner type, the instruction does not ensure full alignment with EUDR requirements.

Figure 12: Procedure for recording GPS coordinates for plantations (source: Instruction 2492/MAF (2020))

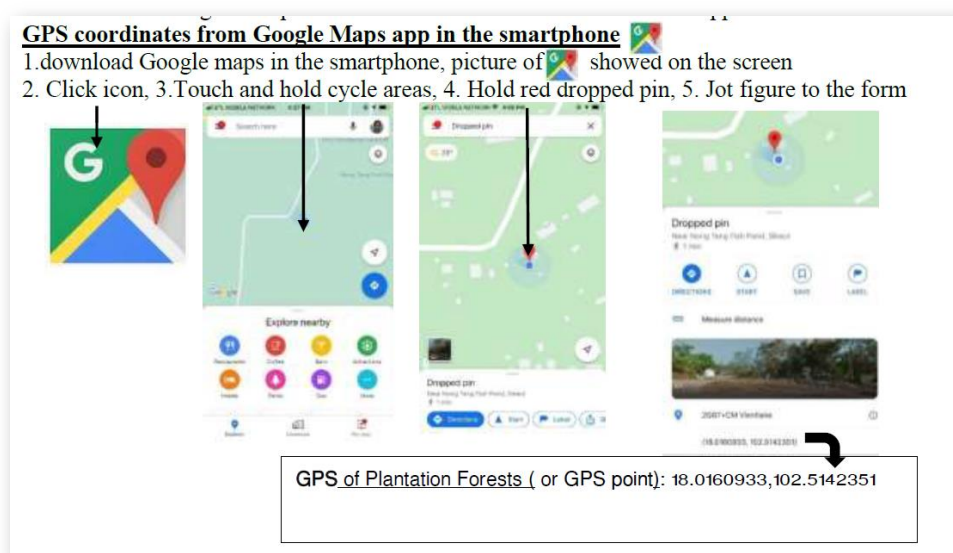


Table 15: Plantation plot registration according to owner type

Owner type	1600 m ² – 5 ha ⁶⁹		>5 ha	
	Single point	Polygon	Single point	Polygon
Individuals	x		x	
Organizations	(x)	(x)	(x)	(x)
Legal Entities		x		x

Coffee traceability systems

According to the interviews, most of the Lao coffee producers such as local smallholder farmers, cooperatives and private companies have been using traditional paper-based recordkeeping systems to record the coffee supply. The Lao companies and cooperatives remain the main interested parties to

⁶⁸ The use of the term ອົງການຈັດຕັ້ງ or (ການຈັດຕັ້ງ in the instruction) is translated with “organization” and can include any organization not organized as a legal entity (including cooperatives, producer groups, associations, yet even villages, government-related entities, etc.).

⁶⁹ 1,600 m² is the size of 1 Rai, a unit of land measurement in Thailand

international markets and therefore are obliged to have traceability systems, logistical capabilities and financial resources to export their coffee to regional and international markets, especially to Asia, Oceania, North America and Europe.

As an example of a paper-based system, one coffee cooperative in the South of Laos⁷⁰ which is working with 450 coffee producers in 19 villages uses paper-based record to collect cherry beans from cooperative members for milling into parchment and processing into green beans (Arabica and robusta) for domestic consumption (90%) and export (10%) to Japan, New Zealand, China and Taiwan. This cooperative creates a form to record the annual coffee supply (see Table 16) and provide a receipt (Figure 13) when making payments to the coffee producers. The information contained in these two records shows a description of the product without HS Code⁷¹, quantity in kilogramme, unit price and amount but does not provide all information as required by Article 9 of EUDR, especially date and time range of production and geolocation coordinates of plots of land of production which is impossible to track the coffees to each plot of land where they were produced.

Table 16. Structure of annual coffee information from coffee members in 2023-2024.

No	Date of supply	Name	Village cluster	Quantity (Kg)	Price LAK	Total
01	15/10/2024	Mr. A	Kapher	500	50,000	2,500,000
02						

In addition, one cooperative⁷² has been using digital traceability systems, such as the Lao program

Figure 13: Receipt.

RECEIPT

Company:

No.....

Phone number:

Date:.....

Payment to: Mr/Mrs

No	Description	Quantity (Kg)	Unit price(LAK)	Amount(LAK)
1.	White coffee beans	500	50,000	2,500,000
2.				
Total				2,500,000

Signature of cashier

Signature of customer

Sabaishop, Excel as well as Google Dashboard, to collect traceability information from coffee producers, especially for quality control and verification of the origin to align their production with consumer demand for sustainably sourced products. However, details could not be shared with the interviewer. The coffee cooperative is located in the South of Laos and produces coffee and buys coffee from members of the cooperative and further processes into green beans. The processing capacity is at the maximum

⁷⁰ Interviewee 19

⁷¹ Harmonized System (HS) codes are commonly used throughout the import and export process for the classification of goods. The Harmonized System is a standardized numerical method of classifying traded products ([Harmonized System \(HS\) Codes](#), retrieved 28/11/2024)

⁷² Interviewee 14

about 8,000 tons/year of cherry beans and 939 tons/year of green beans of which 99% was exported to EU and 1% to Thailand. For traceability, this cooperative has been applying the Fairtrade standard⁷³.

Rubber traceability systems

Traceability for rubber production in Laos is still limited and, if applied, with simple paper-based systems. Many rubber plantations are managed by smallholder farmers who rely on intermediaries for trading with processing companies making traceability system complex and complicated. According to the survey, rubber companies use only simple paper-based forms to record rubber supplies from farmers. A sample is shown in Figure 14 (and the translation of the structure in Table 17) created by Interviewee 27 to record the rubber supply from rubber farmers. This form indicates the village name, the name of farmers, items (Latex), Price/Kg, amount of cash and signature including signature of a note taker, certified person, the cash receiving person and the cashier. It does not indicate the geolocation coordinates where rubber was produced. Most of the rubber is sold as raw latex and exported directly to China and Vietnam. Therefore, there is a need to support the Lao rubber producers to take into consideration such requirements in their supply chain control systems.

Figure 14: Record note for rubber purchase

Table 17: Structure of purchase record for rubber (translated from Figure 14)

Name of village:

Name of farmers	Details of rubber in Kg	Price/Kg	Amount of cash	Signature

Note taker

Certified person

Money receiver

Money payer

⁷³ Fairtrade supports producers to protect forests and prevent [Deforestation](#) (retrieved 28/11/2024)

Cattle traceability systems

The Cattle sector in Laos faces significant challenges in implementing effective traceability systems, with most livestock businesses currently lacking formal mechanisms to track the origin of their products. According to interviews, most livestock operators have only basic traceability systems in place with simple trading records or sales receipts to certify the origin of their cattle, but these documents provide limited details and do not meet the traceability requirements required to track the cattle to their origin

Figure 15. Receipt of sales of animals (and translated structure)

Receipt of sales of animals	
I am the Head of Village:District..... Province	
Have issued this receipt to Mr Khamchanh....Age 56 Occupation..... Unit...01... Dongbang Village.....Saybuly District and Savannakhet Province	
Which sold 1 cow with a calf to Mr. Phim	
Details of characteristics of the cattle	
Head No 1 One cow with a calf	
Dongbang Village	Date 16.9.2024
Sealed and signed by Village Chief	Head of village Unit No 1

or establishment. Figure 15 displays a collected sample form. Information includes the names of seller and buyer and the product specification (one cow with a calf) and the form is sealed and signed by the village chief.

Cattle production in Laos is mostly small-scale and lacks a formal traceability system. Cross-border cattle trade, especially to neighbouring countries, has increased, but weak record-keeping and animal health monitoring pose challenges for traceability. Without established traceability system, it is difficult to verify origin, health status, production and breeding records. Developing a traceability framework for cattle would require a robust infrastructure, training for farmers, and support from both government and private sector stakeholders.

The lack of standardized traceability systems means that it is generally impossible to verify where or how cattle products were raised and processed.

5.2. Readiness for EUDR Traceability Requirements

Overall, traceability in Laos faces significant hurdles but also opportunities for development, especially as global market demands for sustainable and traceable products continue to grow.

The EUDR traceability requirements are outlined in article 9 “Information requirements” of the regulation. In order to demonstrate, that the relevant products fulfil the conditions of (1) being deforestation-free, (2) having been produced in accordance with the relevant legislation of the country of production and (3) being covered by a due diligence statement (article 3), operators are obliged to collect a range of information, documents and data for their due diligence. It must be pointed out that Lao suppliers are not subject of these obligations but collecting this information and make it at disposal of EU clients is a prerequisite for most information requirements, e.g. geolocation data. In addition, as this data can largely only be collected at the first stages of the value chain (e.g. farmer and first producer level), Lao stakeholders who wish to export their products to the EU have an interest in providing the requested information. However, some information requirements of Art. 9 are not requested for the whole Lao supply chain but only for the last actor in the Lao supply chain for which the operators must register the supplier contact information (Art. 9e). Buyer contact documentation (Art. 9f) is an information requirement for operators or traders for any business, operator or trader to whom the relevant products have been supplied and thus not relevant for the Lao supply chain.

The following section summarizes the findings from the interviews and the desk study and categorizes them along the different information requirements.

- **Product description (Art. 9a):**

The majority of participants in the survey include only a basic product description in their records, used commonly in the local context and often not in line with trade names used in the HS classification system. In the case of relevant products that contain or have been made using wood, the common name of the species and their full scientific name is required. This is challenging for natural timber in the Lao context as traceability systems from pre-harvest inventories to processing, often used only local Lao names in the past or are still using them. Yet, the same local names are sometimes used for different species or a species can have different Lao names. This issue has been addressed already under the VPA process as a FLEGT license would have requested the scientific name of a product. The National Agricultural and Forest Research Institute (NAFRI) has compiled and revised existing species information and issued a tree species master list where species have been codified. These codes will be used in the data structure of traceability records compliant with the Lao TLAS.

- **Product quantity documentation (Art. 9b):**

The majority of participants in the survey indicated to document the quantity (kg, volume, net mass) of the products produced and sold. As the price that producers and sellers receive from their buyers is usually based on the weight of the produce, most of the interviewees do some kind of simple – mostly paper based – record keeping. But the records do not contain HS codes of relevant products as listed in Annex I of EUDR. For timber, the supply chain control of the Lao TLAS requires information on product quantity, scientific names and HS codes, yet implementation is currently still weak.

- **Geolocation documentation (Art. 9d):**

One crucial EUDR requirements is the documentation of geolocation of the plot of land where the commodity is grown. Plots of land with less than 4 ha and stalls for cattle require a single geolocation point, while plots larger than 4 ha require the provision of polygons providing information on the shape and size of the plot of land. This information is used in particular to verify the deforestation-free and forest-degradation free statement of the commodity. However,

the interviews showed that this information is still largely unavailable in the agricultural sector in Laos with only 2 companies – one in the coffee and one in the rubber sector – confirming that they have recordings of their plantations geographical information (interview no. 10 & 27). One of the respondents further explained that the company only has the geographic information of its own plantation but is missing information on smallholders who sell their produce to the company (interview no. 27). None of the interviewees from the cattle sector was able to provide the geolocation for stalls or grazing land. Supply chain control systems for natural timber from Production Forest Areas and from conversion areas are locating the origin of logs on paper-based maps, yet with the objective to verify the location during field monitoring and no GPS points are recorded. Batches of plantation timber can be traced back with GPS points and polygons, available on the plantation registration certificate and recorded during application for registration. Yet, the size threshold for point and polygon information is different compared to EUDR. DOFI is currently drafting a decision on the management of confiscated timber, including supply chain control. As briefly mentioned in Chapter 0, the geolocation of the origin of seized and later officially confiscated timber is hardly reproducible and confiscated timber thus not qualified to be exported in compliance with EUDR.

- **Documentation of date and range of production (Art. 9d):**

Together with the requirement of documenting the geolocation of the plot of land where the commodity is grown, data on the date and time range of production is essential in order to assess if a commodity truly is deforestation- and forest degradation-free in terms of the cut-off date of 30 December 2020, defined in the EUDR. During the interviews only 3 respondents – 2 in the coffee and 1 in the rubber sector – indicated that they would be able to provide this documentation when asked by a buyer. However, these interviewees did not provide any further information on what form the documentation takes. Supply chain control forms for PFAs and conversion areas are indicating the date of respective operations (harvesting, transport, scaling/grading, sales). Plantation owners do not need a harvest permit but have to report the number of trees and species before felling to the District Agriculture and Forestry Office. This information will also be registered in the IMS, yet needs to be linked to the sales documentation between plantation owners and buyers. FLEGT-FC will address this in the forthcoming piloting.

- **Supplier contact information (Art. 9e):**

When asked if there is any documentation of supplier contact information, such as name, postal address and email address, 8 respondents – 7 out of them from the coffee sector – confirmed that they would be able to provide information on their suppliers. Natural timber is managed by government entities and sold to sawmills and export of logs is not allowed. Thus, export documents of allowed semi-finished or finished products according to the list of eligible products for export, regularly updated by MAF will have to provide supplier contact information. Plantation timber can be linked to the supplier via the plantation registration certificate which needs to be attached to the sales documentation. The plantation registration certificate does not indicate an email address. As plantation timber can theoretically be exported as round wood to EU markets, this would be a gap.

- **Buyer contact documentation (Art. 9f):**

As stated above, this requirement is not relevant for Lao supply chains but only for business-to-business between an operator and its client. Nevertheless, the following observations have been made when asked if there is record keeping of the name and address of the buyers of the respective products, some respondents to the survey indicated that they document the name and contact information of the buyer but no email addresses are recorded. However, some respondents also explained that they do not keep any record of the buyer, who often takes the

form of a middlemen rather than a producing entity or exporter (interviewee 1). Existing sales document for timber and timber products are in line with buyer contact requirements of art. 9f with exception of email addresses.

- **Transportation records:**

Transport records are not specifically addressed in EUDR, article 9. Nevertheless, in order to be able to trace the commodities over the whole supply chain, transportation records shall ensure that non-compliant commodities may not be mixed with compliant commodities. . However, only few reported to have these records and if they do, it largely did not cover all steps. Most of the respondents referred to their buyers as the responsible entity to provide this information.

5.3. Other Traceability Challenges

Traceability challenges in the timber sector have been intensely discussed in the VPA process and addressed through the amendment of key regulations on supply chain control. Yet, gaps or inconsistencies are still existing in the legal framework (see Table 14) regarding some timber sources and the key regulation for timber processing and trade, 1045/MAF (2023) does not yet cover all Lao TLAS requirements. Across the coffee, rubber and cattle sector in Lao PDR, the stakeholder interviews provided insights into existing traceability practices and gaps. Stakeholders in these agricultural sectors face distinct challenges, with varying levels of traceability system implementation. The following points highlight common challenges identified as well as sector-specific concerns.

Complex Supply Chain Structure

All studied sectors are shaped by a complex supply chain structure, involving smallholder farmers, SMEs, cooperatives and large companies, with each of them having different roles and capabilities. The Lao coffee sector for instance relies on numerous independent smallholders who grow their coffee in small plots of lands before selling it to cooperatives and companies. The farmers often rely on middlemen as they are lacking access to direct market channels, and this is adding another layer of complexity to the supply chain (see for example interview 1, 23). This decentralized model presents logistical hurdles for establishing strict traceability along the chain of custody as required by the EUDR as it complicates the process of tracking origin data down to individual farms.

1. Undocumented trading practices

A large share of agricultural products is traded informally across borders without any records. Such volumes of trade generate only relatively small income which is not subject to taxation under customs procedures, e.g. middlemen collect green coffee beans with a simple record form against cash payment only and bring them across the border to sell to Vietnamese traders. Practices like these make it nearly impossible to track the coffee beans to their geolocation of production. Another aspect of concern is the export of unprocessed commodities. For instance, the Lao legal framework prohibits the trading of unprocessed rubber, due to a lack of processing industries in Laos, however, unprocessed rubber in many provinces has been sold to neighbouring countries with only simple records.

2. Data Collection and Management

A key issue is the lack of digital traceability tools, with most producers and manufacturers relying on paper-based records that do not align with requirements of international regulations, such as the European Union Deforestation Regulation (EUDR). The lack of systematic data management hinders transparent and effective traceability, posing risks to market entry, particularly in regions with strict regulatory requirements. In the Lao coffee sector, record-keeping mainly includes data on the quantity and quality of coffee collected, while geolocation and detailed farming practices are largely

undocumented with exception of some companies with niche products for export markets but which are currently focusing on a more coarse indication of the origin (members, villages) and not the plot of production. Similarly, trade documents in the cattle sector often provide only basic data such as the owner of the cattle and the characteristics of the livestock. This gap in information could pose issues when exporting to the EU, especially for smallholders who lack the infrastructure or training for comprehensive data collection.

Additionally, Lao regions that are remote or less connected can face logistical barriers in tracing products effectively from farm to consumer. This can impact the accuracy and timeliness of traceability information.

3. Resource and Capacity Limitations

Many producers in Laos, particularly smallholders, operate with limited financial resources and technical skills and knowledge (interview no. 6, 17). Implementing the technology and processes required for traceability involves substantial upfront costs, placing a strain on resources. Furthermore, a shortage of resources, such as access to mobile apps or traceability software, is hindering the ability of producers to meet traceability requirements. In addition to financial and technological resources, many farmers are lacking the capacity to gather the requested information and managing the associated paperwork. Training farmers and cooperative members to manage traceability effectively will require support and investment in capacity-building efforts.

4. Coordination on traceability within Lao PDR

An interviewee mentioned the limited support from local institutions or the government in terms of policies or subsidies to help smaller farms adapting or establishing a traceability system aligned with international requirements such as the EUDR (interview no. 12).

5. No request from buyers for documentation

Many stakeholders report that the main reason for not having a comprehensive traceability is that neither their current buyers nor government institutions are asking for it. Hence, they do not see the need to develop and implement such a system which for them is mostly a cost and capacity question in the first place (interview no. 2, 4, 8, 12, 24, 25).

5.4. Potential Benefits of Traceability Systems

The review on current traceability systems is concluded with an outlook on potential benefits of traceability systems, aligning with EUDR traceability requirements. The outlook is based on findings of the global traceability study (GIZ, 2024) and complemented by observations during the implementation of this study and applies in the Lao context.

1. Market Access and Competitive Advantage

Aligning with EUDR traceability requirements could position Lao coffee favourably in the EU market, providing access to high-demand markets with sustainable sourcing requirements. By adopting traceability systems, Lao coffee could be marketed as a premium, environmentally conscious product, appealing to sustainability-focused consumers and businesses across Europe.

2. Improved Supply Chain Efficiency

Implementing traceability measures can streamline the supply chain, fostering a more organized structure from production to export. For example, tracking production from the cooperative level to final export can improve efficiency and oversight, reducing waste and enhancing quality control. This level of

organization can benefit both large-scale companies and cooperatives by improving product consistency and ensuring a reliable supply chain.

3. Enhanced Farmer Income and Stability

Enhanced traceability on deforestation-free production could lead to better income stability for Lao producers especially for local farmers and small households via enhanced traceability along the chain of custody, producers can access markets with higher price points, leading to potential income improvements. This increased revenue stream could also support funding initiatives for smallholders, further stabilizing the industry. Additionally, it could incentivize cooperatives and large producers to offer advanced payment systems and financial support for farmers, creating more reliable income for smallholders.

4. Support for Sustainable Agricultural Practices

EUDR traceability requirements align with international and national sustainability objectives by encouraging practices that prevent deforestation. Adopting these practices not only benefits the environment but also improves soil health, yield sustainability, and resilience against climate-related risks. As companies expand into sustainability-focused practices, such as supporting nurseries and promoting crop diversification, they can foster long-term environmental and economic benefits for the coffee, rubber, cattle and timber sector and local ecosystems.

In conclusion, while ensuring traceability from the plot along the supply chain poses challenges for the Lao agricultural sector and at a lower level for the timber sector, due to logistical complexities, resource limitations, and data management needs, it also offers significant advantages, especially in terms of market access, economic stability, and sustainable development. Establishing traceability solutions will require collaboration between cooperatives, government agencies, and international stakeholders, but it presents an opportunity to enhance the sector's global competitiveness and resilience.

6. Conclusions

This study provides an overview and analysis of the current status of traceability systems within the supply chains of the Lao coffee, cattle, rubber, and timber sectors. The findings reveal significant gaps when compared to the EUDR's criteria for traceability for coffee, cattle and rubber and some remaining gaps for timber.

While a large number of stakeholders interviewed reported to use some kind of traceability system, most of them only cover single steps of the supply chain rather than allowing for tracing an end-product back to the plot of land where it was sourced from. When comparing the current state-of the art of data collection with the EUDR requirements for traceability as outlined in Article 9 of the regulation, most of the respondents were only able to provide information on some selected criteria such as product quantity or buyer information.

The timber sector can be considered the most advanced sector when it comes to traceability, resulting from past achievements such as the (ongoing) TLAS development and related trainings as part of the FLEGT VPA negotiations. However, challenges remain due to a complex distribution of roles and overlapping responsibilities, a regulatory framework which is still under development. Similarly, the coffee sector appears to be relatively advanced in comparison to other sectors, given the presence of a number of certification schemes in the sector as well as existing export links to markets with high entry requirements such as the EU. The dominance of independent smallholders poses a challenge to traceability to the plot of production; however, the increasing role of cooperatives may positively impact

this hurdle. In addition, stakeholders from this sector seemed particularly interested in increasing their EUDR readiness through an adaptation of production practices and capacity building. The Lao rubber sector shows varying levels of traceability maturity, with some established practices but limited alignment with the EUDR's strict traceability requirements, particularly in geolocation data and deforestation-free production. The cattle sector, however, faces more substantial challenges due to less-developed systems, inconsistent or non-existent record-keeping, and a lack of integration across supply chain actors.

These deficiencies underscore the need for sector-specific interventions to strengthen traceability infrastructure, enhance producer awareness, and establish cross-sector coordination mechanisms.

The present study sheds light on the current implementation of traceability systems for four relevant commodities and products under the EUDR to assess the readiness of Lao supply chain actors to gather and provide information. Due to the importance of smallholders for the production of commodities and the existence of middlemen, often acting informally, the tracking of data from smallholders to first level producers remains to constitute a huge challenge for the so-called “first-mile traceability”. The study has not directly assessed options for applicable traceability solutions in the Lao context but is linked to a global study (GIZ, 2024) which provides a “step-to-step-guide” on which aspects to keep in mind for establishing inclusive traceability solutions aligned with the EUDR. Details can be found in the final report of the global study

7. Recommendations

Lao stakeholders who are exporting or wish to export their products from relevant commodities to EU markets need to prepare for and provide reliable traceability information to operators who must exercise due diligence under the EUDR. To achieve this, we recommended to develop traceability systems for Lao producers in a multi-stakeholder approach and build capacity to demonstrate that their products are deforestation-free and forest degradation-free.

1. *Engagement of Lao key stakeholders*

- To achieve traceability solutions, all actors along the supply chains will need to actively participate in the further development of traceability solutions and understand their responsibilities. Importantly, policy decision makers play critical roles to promote an enabling environment to support the implementation of these solutions.
- There are important lessons learned from the VPA process and the development of a TLAS for the other sectors. The legal framework for supply chain management needs to be duly assessed regarding gaps and inconsistencies and reflected with the current practice. A multi-stakeholder process will help to identify challenges, develop tailored solutions, increase awareness and promote a better and faster uptake of solutions by stakeholders. This is best organized sector by sector with the involvement of relevant government agencies, private sector and civil society.
- Potential solutions shall include an assessment of the appropriateness of existing options summarized by the global study. For this endeavor, the global study has developed a “Step-to-step guide to select a fit-for-purpose traceability solution” which is annexed to the report. The guide is targeted at users in countries of production who wish to select or adapt a traceability solution fit for their purpose. Potential solutions shall also embrace broader sustainability goals and favor inclusive traceability solutions.

- Traceability and transparency are important elements for implementing EUDR. Coordination among sectors and development of a national action plan on concrete next steps for establishing traceability system for relevant commodities and products under EUDR would allow synergies and exchange of lessons learned among sectors.

2. *Technical and financial support for development of traceability system*

- Once stakeholders agreed on solutions, financial support must be provided by development partners to develop and invest in a comprehensive traceability system. The traceability system shall empower all related actors along the supply chain to register and record their production information and thereby keeping the specific due diligence reference number to prove their products (commodities) are deforestation-free and forest degradation-free. TLAS development should consider taking into account the traceability requirement of EUDR.
- The traceability systems need to be robust by reflecting the local context and capacity. For remote areas, paper-based data recording may be appropriate if IT solutions are proving to be too cost-intensive or technically not manageable by smallholders.⁷⁴
- To ensure effective and efficient implementation of the system, IT infrastructure and technical equipment such as computers, barcode scanners, QR code systems, GPS devices for information and geolocation collection of plots of land need to be provided to smallholders or local cooperatives who lack financial means to buy them.
- Develop a simple manual with step-by-step procedures (App) for Lao producers to effectively collect traceability information. The information to be collected and provided shall include (Article 9 of EUDR):
 - The trade name and types of relevant products;
 - In the case of relevant products that contain or have been made using wood, the common name of the species and their full scientific name;
 - Quantity in kilograms or quantity in net mass, volumes or number of items;
 - Country of production (Laos);
 - Geolocation of all plots of land where the commodities were produced; for plots of land of more than four hectares used for the production of the relevant commodities other than cattle, this shall be provided using polygons with sufficient latitude and longitude points to describe the perimeter of each plot of land;
 - Date or time range of production, HS codes, including the name, postal and email address of the supplier to the EU operator.
- Create and establish an expert group which consist of related public sector, private sector and CSOs to provide technical support and services to Lao producers as required.

3. *Capacity building*

- Provide trainings on information collection to upstream producers such as Lao smallholder farmers, cooperatives and companies to strengthen their capacity to be ready to collect and give accurate information to their buyers. The trainings would include steps and procedures of data collection, use of digital tools and software to collect traceability information.
- Conduct awareness raising activities on the benefits of traceability systems through online trainings and other means to ensure the understanding and learning by related producers.

⁷⁴ Smallholders are not supposed to bear the cost of traceability systems, but operators may offer cost-sharing, capacity building free of charge or investments as part of their risk mitigation measures.

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Annex B: Short Profile of Interviewed Stakeholders

No	Region	Commodity	Company Type	Traceability System	Markets	Stakeholder Description
1	Central	Cattle	Producer	Paper-based	Domestic market	Stakeholder 1 is a live cattle production company with over 200 buffaloes and over 20 cows. The live cattle are sold to middlemen.
2	Central	Cattle	Producer	Not available	Domestic market	Stakeholder 2 is a live cattle production company breeding their own cattle. The live cattle are sold to middlemen.
3	Central	Cattle	Producer	Paper-based	Domestic market	Stakeholder 3 is a live cattle production company breeding their own cattle and buying from local farmers. The company owns a slaughterhouse and is selling meat domestically.
4	Central	Cattle	Producer	Not available	Domestic market	Stakeholder 4 is a production company breeding cattle and goats. The company sells live cattle to middlemen and meat on the domestic market.
5	North	Cattle	Producer	Paper-based	Domestic market	Stakeholder 5 is a production company breeding their own cattle and buying from local farmers. The live cattle are sold to middlemen.
6	Central	Cattle	Producer	Not available	Domestic market	Stakeholder 6 is a production company breeding their own cattle. The live cattle are sold to middlemen.
7	Central	Coffee	Processor	Paper-based	EU, domestic market	Stakeholder 7 is a coffee processor, mainly processing coffee beans collected from local farmers in Champasak and Xekong Province. The company is producing 40-50 tons of coffee per year on average.
8	Central	Coffee	Processor	Paper-based	Domestic market, EU, USA	Stakeholder 8 is a coffee processor, processing beans collected from local farmers/farmer's cooperatives in Champasak province. The company is producing 30 tons of roasted coffee per year on average.

9	North	Coffee	Processor / Producer	Paper-based	Japan, USA, Germany	Stakeholder 9 is a coffee processor, processing beans collected from local farmers as well as their own plantation. The company is producing up to 80 tons of coffee per year. Main operations of stakeholder 9 encompass the entire coffee production process, starting with supporting coffee tree nurseries for local villagers to processing and roasting the coffee beans.
10	South	Coffee	Processor / Producer	Digital system	Japan, Korea, USA, EU	Stakeholder 10 is a coffee processor, processing beans collected from local farmers as well as their own plantation. The company is producing about 30 tons of green beans per year and 12 tons of roasted beans per year on average. Main operations of stakeholder 10 encompass providing technical and financial support to local farmers as well as processing and roasting coffee beans.
11	Central	Coffee	Association	Not applicable	Not applicable	Stakeholder 11 unites various private sector participants involved in the coffee industry, including producers, processors, operators, and exporters. Currently, the association comprises 67 private sector companies and 75 cooperatives. By bringing together these stakeholders, the association fosters collaboration across all levels of production and export, creating a strong network to support the growth and development of the industry on practical but also policy level.
12	Central	Coffee	NGO	Not applicable	Not applicable	The NGO's program seeks to enhance income diversification for producers while fostering gender equality within the tea and coffee value chains. Key objectives include strengthening research and education in coffee production, with a particular focus on coffee breeding, as well as developing educational tools to improve knowledge among producers and entrepreneurs.

						The program prioritizes boosting coffee production across seven northern provinces, which serve as a central hub for these activities. Additional efforts include supporting the Lao coffee sector through branding initiatives and facilitating the country's membership in the World Coffee Organization, thereby expanding export opportunities to the European Union market.
13	South	Coffee	Producer / Cooperative	Paper-based	Domestic and international market	Stakeholder 13 is a coffee producer and cooperative, processing the beans collected from the coffee cooperative's members. The production volume is up to 200 tons per year on average. Main operations of stakeholder 13 encompass the provision of trainings and financial assistance to farmers to increase productivity, supporting the development of coffee plant nurseries and expanding market opportunities by participating in coffee exhibitions and fostering market connections.
14	South	Coffee	Producer / Cooperative	Digital system	EU, Thailand	Stakeholder 14 is a coffee producer and cooperative, processing the beans collected from the coffee cooperative's members. The production volume is 5,000 tons per year on average. Main operations of stakeholder 14 encompass targeted trainings and capacity building, for instance on organic fertilizer production, facilitating coffee export, as well as the processing of coffee beans.
15	South	Coffee	Processor	Paper-based	EU, USA	Stakeholder 15 is a coffee processor, processing beans collected from small and medium enterprises and local farmers. The production volume of the past years was between 5,000 to 6,000 tons but decreased to 1,000 tons per year on average after Covid-19.
16	South	Coffee	Processor / Producer	Paper-based and digital	Domestic market, Japan, Thailand	Stakeholder 16 is a coffee processor, processing beans collected from local farmers (not in 2024) as

						well as their own plantation. The production volume is between 4,000 and 5,000 tons per year on average.
17	South	Coffee	Processor / Producer	Paper-based and digital	Denmark, Finland, Thailand	Stakeholder 17 is a coffee processor, processing beans collected mainly from its own plantation as well as local farmers who are also part of cooperatives. The production volume are 300 tons of green bean per year on average. The green beans are sent to Europe for further processing.
18	South	Coffee	Producer / Cooperative	Paper-based and digital	Japan, Zealand, Taiwan, New China,	Stakeholder 18 is a coffee producer and cooperative, processing beans collected from their cooperative's members. The production volume are about 1,500 tons of roasted coffee per year on average. The cooperative consists of 19 village groups with 450 members located in Paksong District. They have an EU-organic certification as well as an (expired) Fair Trade certification.
19	South	Coffee	Producer / Cooperative	Paper-based and digital	Domestic market	Stakeholder 19 is a coffee producer and cooperative, processing beans collected from their cooperative's members who use agroforestry systems and their own plantation. The production volume are about 200 tons per years on average. For further processing, the producer is selling the coffee beans to larger companies.
20	South	Coffee	Processor / Producer	Paper-based and digital	Thailand, Japan, EU	Stakeholder 20 is a coffee processor, processing coffee from their own plantation, which in part is under an agroforestry system, as well as local farmers. The production volume is about 1,400 tons of green bean per year on average. For further processing, the coffee is sold to international buyers. Stakeholder 20 holds a Rainforest Alliance and Café Practices Certificate.
21	South	Coffee	Processor	Paper-based	Domestic market, Vietnam, Thailand	Stakeholder 21 is a coffee processor, largely processing coffee from local farmers. The production volume is about 10,000 tons per year

						on average. Some coffee is exported as green beans for further processing to international buyers while some is roasted in their own processing factory.
22	South	Coffee	Processor / Producer	Paper-based	Domestic market, Thailand, China	Stakeholder 22 is a coffee processor, largely processing coffee collected from local farmers while only a small share comes from their own plantation. The production volume is up to 3,600 tons per year. They produce roasted beans for the domestic market but also sell green beans to international buyers for further processing.
23	North	Coffee	Processor	Not available	Domestic market	Stakeholder 23 is a coffee processor, processing coffee collected from local farmers. The production volume is 4 tons of green beans per year on average. Main activities include training, capacity building and financial assistance to farmers. The beans are sold to a domestic processor for roasting.
24	Central	Rubber	Association	Not applicable	Not applicable	Stakeholder 24 is a rubber association with 30 members who produce an average of 400,000 tons of rubber per year. Main activities include training and capacity building for members as well as representation of interest of its members.
25	Central	Rubber	Producer	Paper-based	Domestic market	Stakeholder 25 is a rubber producer, who has its own rubber plantation. The raw rubber harvested from these plantations is sold to a domestic semi-processing facility. The production volume is 630 tons per year on average.
26	Central	Rubber	Processor	Paper-based	Vietnam	Stakeholder 26 is a rubber processor who sources the rubber from local farmers only. After collection by mid-traders, the company processes the rubber into rubber blocks in "SVR10" form. The production volume is 54,000 tons per year on average.

27	North	Rubber	Processor / Producer	Paper-based	China	Stakeholder 27 is a rubber producer who sources its rubber from local farmers as well as its own plantation. The production volume is 20,000 tons per year on average. Currently, the company only exports raw latex in the form of cup rubber.
28	North	Rubber	Processor / Producer	Paper-based	China	Stakeholder 28 is a rubber processor who sources its rubber from local farmers as well as its own plantation. The production volume is 38,000 tons per year on average. The rubber is further processed before export.

Annex C: Onsite Stakeholders Interview Plan

Date (MDY)	Time of Int	Region	Province
Monday, September 23, 2024	08:00-09:00	Central	Vientiane Capital
Monday, September 23, 2024	11:00-12:00	Central	Vientiane Capital
Monday, September 23, 2024	1:30:00-14:30	Central	Vientiane Capital
Tuesday, September 24, 2024	13:00-14:00	Central	Borikhamxay
Wednesday, September 25, 2024	1:30:00-14:30	Central	<i>Borikhamxay</i>
Wednesday, September 25, 2024	16:30:00-17:30	Central	Borikhamxay
Thursday, September 26, 2024	1:00:00-14:00	Central	Vientiane Capital
Thursday, September 26, 2024	09:00-10:00	Central	Vientiane Capital
Friday, September 27, 2024	9:00-10:00	North	Luang Prabang
Friday, September 27, 2024	11:00-12:00	South	Champasak
Friday, September 27, 2024	13:00-14:00	Central	Champasak
Friday, September 27, 2024	17:00-18:00	Central	Vientiane Capital
Saturday, September 28, 2024	08:00-09:00	Central	Vientiane Capital
Monday, September 30, 2024	11:00-12:00	North	Luang Prabang
Wednesday, October 2, 2024	09:30-10:30	North	Xiengkhuang(Afternoon)
Friday, October 4, 2024	13:00-14:00	Central	Vientiane Capital
Saturday, October 5, 2024	11:00-12:00	South	Salavan
Sunday, October 6, 2024	09:30-11:30	South	Champasak
Monday, October 7, 2024	09:00-10:00	South	Champasak
Monday, October 7, 2024	15:30-16:30	South	Champasak
Tuesday, October 8, 2024	11:00-12:00	South	Champasak
Tuesday, October 8, 2024	14:00-15:00	South	Champasak
Wednesday, October 9, 2024	08:30-09:30	South	Champasak
Wednesday, October 9, 2024	13:00-13:30	South	Champasak
Friday, October 11, 2024	13:00-14:00	South	Champasak
Monday, October 14, 2024	09:00-10:00	South	Champasak
Monday, October 14, 2024	11:00-12:00	South	Champasak
Wednesday, October 16, 2024	Submitted by email	North	Luang NamTha
Monday, October 28, 2024	Submitted by email	Central- North	VTE-LNTH

Annex D : Questionnaire

Questionnaire for EUDR Lao Traceability Study

INTRODUCTION

- **Context:** The EU Delegation to Lao PDR has launched this study on traceability to assess the national context of relevant Lao commodities (cattle, coffee, rubber and timber) to adapt to the EU Deforestation-free Regulation's (EUDR) requirements. The study is part of the EUDR Engagement Project with Lao PDR to inform on the EU approach to reducing EU-driven deforestation and forest degradation, the impacts of climate change and biodiversity loss. Under this study, we are particularly interested in information on the structure of relevant supply chains as well as existing national traceability systems.
- **Outlook for the interview:** For this reason, we would like to learn more about how the structure of the Commodities supply chain in Laos is structured and what methods are currently used by farmers, processors, traders and exporters to document the supply chain. We are not only interested in advanced traceability tools but also simple ways of documentation that stakeholders in this supply chain use.
- **Consent for data protection and recording:** You have received the data protection information in advance. I would like to ask you again whether you agree to the recording of the information as well as to the data protection regulations.

Section 1: General Information

This section collects basic information about the respondent and their business.

1.1. Name of Respondent:

- Name: _____
- Position: _____
- Phone number: _____
- Email: _____

1.2. Company/Organization:

- Name: _____
- Type of Operation:
 - ☐ Company
 - ☐ Cooperative
 - ☐ Association
 - ☐ International Entity

- ☐ Other: _____
- Office location (Province/District/Village): _____

- Production location (Province/district/village) : _____

- Main Commodity
 - ☐ Coffee
 - ☐ Rubber
 - ☐ Cattle
 - ☐ Timber

2. Main activities

Please briefly describe the main activities of your organization/association.

- What is the primary focus of your business?
 - ☐ Production (Smallholder)
 - ☐ Production (Concession)
 - ☐ Trading
 - ☐ Processing
 - ☐ Export
 - ☐ Other: _____
- How many years has your business been operating?

- How many tons/year (or another unit) is your business producing?

- (For Processor & Exporter) Do you export your products?
 - ☐ Yes
 - ☐ No
- If yes, what are your export markets (EU, Countries)? : _____

3. Certification

Is your organization participating in a voluntary certification scheme? Please specify

Section 2: Supply Chains

2.1 Please give me a brief description of how your commodities supply chain is structured, i.e. where and in what form the commodities is grown and which stations it passes through on its way to the end product.

- What is the location of cultivation/production?
- In what form are the commodities grown? (Agroforestry, monoculture, mixed culture, ...)
- By whom are the plantations managed? (smallholders, employees, ...)
- Who are your suppliers? (e.g., farmers, smallholders, traders)
- Who are your buyers? (e.g., producers, traders, processors, exporters)?
- Are there contract farming or formal agreements between different actors?
- In what form is the commodities sold?

Section 3: Traceability System/Record

In this section we would like to gain more insights into if and how you document the different steps of your supply chain. We refer to this documentation as “traceability system”. A traceability system can be a very simple form of paper-based or digital documentation but also a more advanced technological solution to track the different stations of a commodity, such as blockchain.

1. Do you have a traceability system to track your commodities?

- ☐ Yes (fill in question 2, 3 & 4)
☐ No (please go directly to question 5)

2. If yes, What kind of traceability system do you use?

- ☐ Paper-based record (manuals)
- ☐ Digital system (Specify software used):

- ☐ Other (Specify):

3. Please describe how you collect and store information about your production.

- Do you document the quantity (kg, volume, net mass) of the products produced? How do you do that? (*Art. 9b*)
- Do you document the name and address of buyers of your products? How do you do that? (*Art. 9c*)
- Do you document the geolocation of the plots of land where the commodity was grown? How do you do that? (*Art. 9d*)
- Do you document the date or time range of production of a certain product/commodity? How do you do that? (*Art. 9d*)
- Do you document the name and address of suppliers (farmers/producers) of raw materials/products? How do you do that? (*Art. 9d*)
- Do you document the transportation records of your commodities? How?
- For exporter: How do you document the export license?
- For cattle: Do you document where the cattle is kept? Do you document where the cattle is grazing? (*Art. 9d*)

- 4. What are the main challenges for your organization/cooperative/association to collect and document traceability data?** (e.g. cost of implementation, lack of technical knowledge or expertise, resistance from farmers, lack of infrastructure (e.g., internet, electricity), complexity of managing data from multiple farms, ...)

- 5. If no documentation is taking place: What are the main reasons/barriers preventing, you from recording traceability data?**
(e.g. not requested by traders, difficult to collect information, limited capacities)

Section 4: Support and Future Needs

This section gathers information about the support that companies need and their future regarding traceability.

1. What support do you need to implement/improve traceability system to export your products to EU under the EUDR?

(e.g. Training on traceability, financial support, access to technology/software, Government or policy support, etc.)

Section 5: Additional Comments

This section allows respondents to provide any additional feedback or suggestions.

1. Do you have any suggestions or recommendations for improving traceability in your sector (Coffee, Rubber, Cattle, Timber)?

Outlook

We are currently in the phase of conducting interviews with stakeholders from different sectors. After that, we will conduct the analysis and do some more desk study. The study results will be presented during a multi-stakeholder workshop end of November 2024.

Final Note

Thank you for participating in the EUDR Lao Traceability Study. Your feedback is valuable and will contribute to our understanding on where support to enhance traceability in Laos is needed.

Annex E: Stakeholder Selection Process

The selection of stakeholders was made based on a draft list provided by the respective sector associations for coffee and rubber. For the cattle sector, the selection was made based on a list from the Department of Livestock and Fisheries under MAF. During the study it revealed that the Lao Brahman Breeders Association (LBBA) – established in 2022 and endorsed by the Lao Government - represents the stakeholders within the cattle sector. However, the association was not included in the interviews, as its existence was only brought to the research team’s attention during an EUDR outreach workshop to the cattle private sector in Vang Vieng after the interview period.

The selection process was guided by two main criteria: (1) geographic representation across three regions, (2) the availability of stakeholders. Respondents were contacted to confirm their participation, and if unavailable, alternatives were drawn from the pre-existing lists. Government and development partners were excluded from interviews to maintain a focus on those directly involved in the practical implementation of the commodity supply chain. As noted in the introduction, the legal framework on the traceability is not within the scope of this study. Out of 40 contacted stakeholders, 28 interviewees were interviewed, with a response ratio from coffee sector being slightly higher (approximately 75% compared to around 60% for other two sectors), indicating the higher interest of the sector to export its products to EU markets. The following on **Error! Reference source not found.** summarizes the selection process of stakeholders selected for the interview.

Table 18: Number of the Interviewees

Commodities	Type of Operation	Request for interview	Confirmed
Cattle	Companies	10	6
	Total	10	6
Coffee	Association	1	1
	International entity	1	1
	Companies	16	11
	Cooperatives	4	4
	Total	22	17
Rubber	Companies	7	4
	Association	1	1
	Total	8	5
TOTAL		40	28

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